

Unit 1 AP Topics 1-4

1.  Let f be the function given by $f(x) = 2xe^{2x}$.

(a) Find $\lim_{x \rightarrow -\infty} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$.

(b) Find the absolute minimum value of f . Justify that your answer is an absolute minimum.

(c) What is the range of f ?

(d) Consider the family of functions defined by $y = bxe^{bx}$, where b is a nonzero constant. Show that the absolute minimum value of bxe^{bx} is the same for all nonzero values of b .

Part A

1 point is earned for 0 as $x \rightarrow -\infty$

1 point is earned for ∞ or DNE as $x \rightarrow \infty$

$$\lim_{x \rightarrow -\infty} 2xe^{2x} = 0$$

$$\lim_{x \rightarrow \infty} 2xe^{2x} = \infty \text{ or DNE}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for 0 as $x \rightarrow -\infty$

1 point is earned for ∞ or DNE as $x \rightarrow \infty$

$$\lim_{x \rightarrow -\infty} 2xe^{2x} = 0$$

$$\lim_{x \rightarrow \infty} 2xe^{2x} = \infty \text{ or DNE}$$

Part B

1 point is earned if solves $f'(x) = 0$

1 point is earned if evaluates f at student's critical point

0/1 if not local minimum from student's derivative

1 point is earned if justifies absolute minimum value

0/1 for a local argument

0/1 without explicit symbolic derivative

Unit 1 AP Topics 1-4

Note: 0/3 if no absolute minimum based on student's derivative

$$f'(x) = 2e^{2x} + 2x \cdot 2 \cdot e^{2x} = 2e^{2x}(1 + 2x) = 0$$

if $x = -1/2$

$$f(-1/2) = -1/e \text{ or } -0.368 \text{ or } -0.367$$

$-1/e$ is an absolute minimum value because:

(i) $f'(x) < 0$ for all $x < -1/2$ and $f'(x) > 0$ for all $x > -1/2$

-or-

(ii)
$$\begin{array}{c} f'(x) \quad - \quad + \\ \hline -1/2 \end{array}$$

and $x = -1/2$ is the only critical number



0	1	2	3
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The student response earns all of the following points:

1 point is earned if solves $f'(x) = 0$

1 point is earned if evaluates f at student's critical point

0/1 if not local minimum from student's derivative

1 point is earned if justifies absolute minimum value

0/1 for a local argument

0/1 without explicit symbolic derivative

Note: 0/3 if no absolute minimum based on student's derivative

$$f'(x) = 2e^{2x} + 2x \cdot 2 \cdot e^{2x} = 2e^{2x}(1 + 2x) = 0$$

if $x = -1/2$

$$f(-1/2) = -1/e \text{ or } -0.368 \text{ or } -0.367$$

$-1/e$ is an absolute minimum value because:

(i) $f'(x) < 0$ for all $x < -1/2$ and $f'(x) > 0$ for all $x > -1/2$

-or-

Unit 1 AP Topics 1-4

(ii) $f'(x)$ $\frac{-}{+}$
 $\frac{-1/2}{-1/2}$

and $x = -1/2$ is the only critical number

Part C

1 point is earned for the answer

Note: must include the left-hand endpoint; exclude the right-hand “endpoint”

Range of $f = [-1/e, \infty)$ or $[-0.367, \infty)$ or $[-0.368, \infty)$



0	1
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The student response earns all of the following points:

1 point is earned for the answer

Note: must include the left-hand endpoint; exclude the right-hand “endpoint”

Range of $f = [-1/e, \infty)$ or $[-0.367, \infty)$ or $[-0.368, \infty)$

Part D

1 point is earned if sets $y' = be^{bx}(1 + bx) = 0$

1 point is earned for solving student's $y' = 0$

1 point is earned if evaluates y at a critical number and gets a value independent of b

Note: 0/3 if only considering specific values of b

$$y' = be^{bx} + b^2xe^{bx} = be^{bx}(1 + bx) = 0$$

if $x = -1/b$

At $x = -1/b$, $y = -1/e$

y has an absolute minimum value of $-1/e$ for all nonzero b



0	1	2	3
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Unit 1 AP Topics 1-4

The student response earns all of the following points:

1 point is earned if sets $y' = be^{bx}(1 + bx) = 0$

1 point is earned for solving student's $y' = 0$

1 point is earned if evaluates y at a critical number and gets a value independent of b

Note: 0/3 if only considering specific values of b

$$y' = be^{bx} + b^2xe^{bx} = be^{bx}(1 + bx) = 0$$

$$\text{if } x = -1/b$$

$$\text{At } x = -1/b, y = -1/e$$

y has an absolute minimum value of $-1/e$ for all nonzero b

Unit 1 AP Topics 1-4

2. Let f be the function given by $f(x) = 2xe^{2x}$.



Find $\lim_{x \rightarrow -\infty} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$.

Part A

1 point is earned for 0 as $x \rightarrow -\infty$

1 point is earned for ∞ or DNE as $x \rightarrow \infty$

$$\lim_{x \rightarrow -\infty} 2xe^{2x} = 0$$

$$\lim_{x \rightarrow \infty} 2xe^{2x} = \infty \text{ or DNE}$$



0	1	2
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The student response earns two of the following points:

1 point is earned for 0 as $x \rightarrow -\infty$

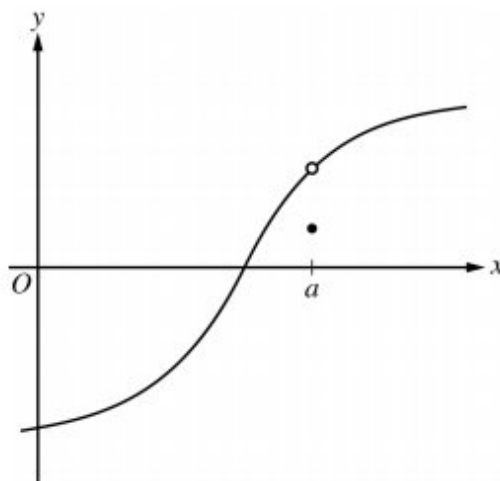
1 point is earned for ∞ or DNE as $x \rightarrow \infty$

$$\lim_{x \rightarrow -\infty} 2xe^{2x} = 0$$

$$\lim_{x \rightarrow \infty} 2xe^{2x} = \infty \text{ or DNE}$$

Unit 1 AP Topics 1-4

3.

Graph of f

The graph of $y = f(x)$ is shown above.

Which of the following is true?

- (A) $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists.
- (B) $\lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x)$
- (C) $\lim_{x \rightarrow a} f(x) \neq f(a)$
- (D) $\lim_{x \rightarrow a} f(x)$ does not exist.



4. If $\lim_{x \rightarrow 3} f(x) = 7$, which of the following must be true?

- I. f is continuous at $x = 3$
- II. f is differentiable at $x = 3$
- III. $f(3) = 7$

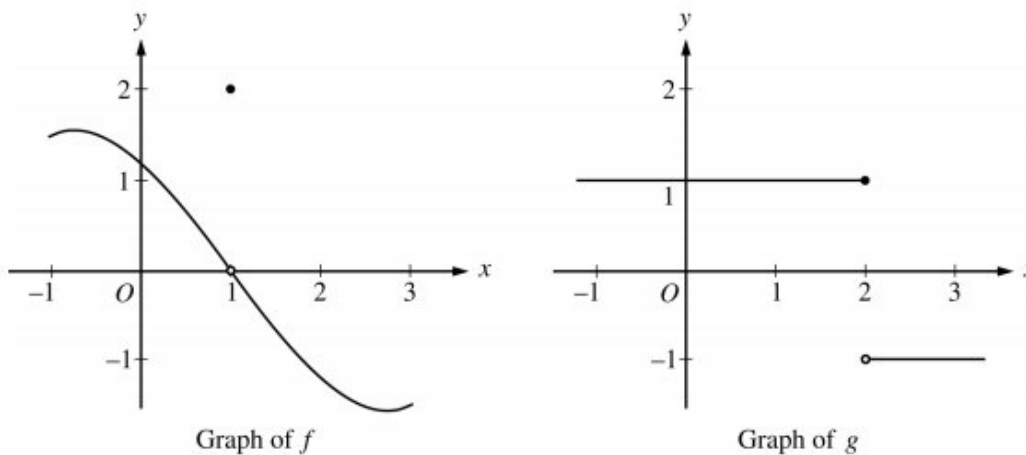
- (A) None



- (B) II only
- (C) III only
- (D) I and III only
- (E) I, II, and III

Unit 1 AP Topics 1-4

5.



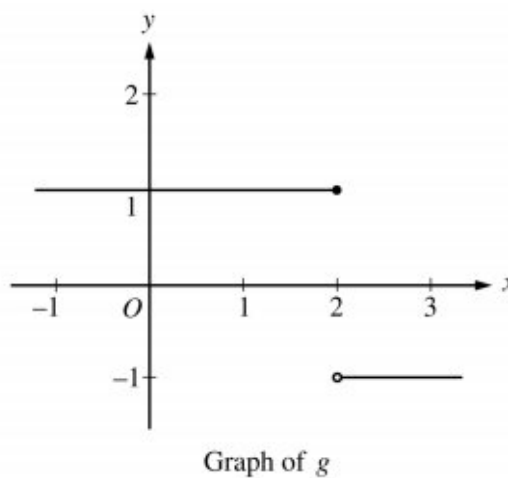
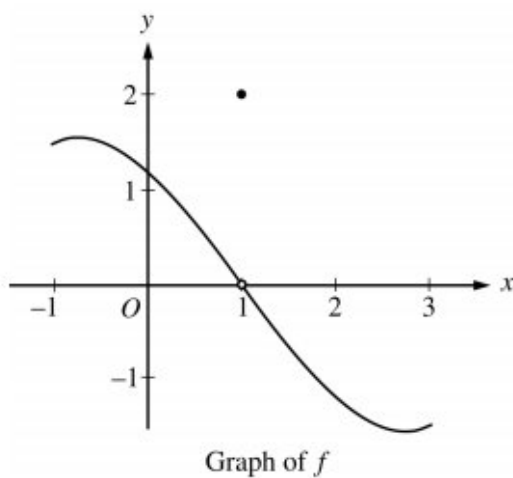
The graphs of the functions f and g are shown in the figures above.

Which of the following statements is false?

- (A) $\lim_{x \rightarrow 1} f(x) = 0$
- (B) $\lim_{x \rightarrow 2} g(x)$ does not exist.
- (C) $\lim_{x \rightarrow 1} (f(x)g(x+1))$ does not exist. ✓
- (D) $\lim_{x \rightarrow 1} (f(x+1)g(x))$ exists.

Unit 1 AP Topics 1-4

6.



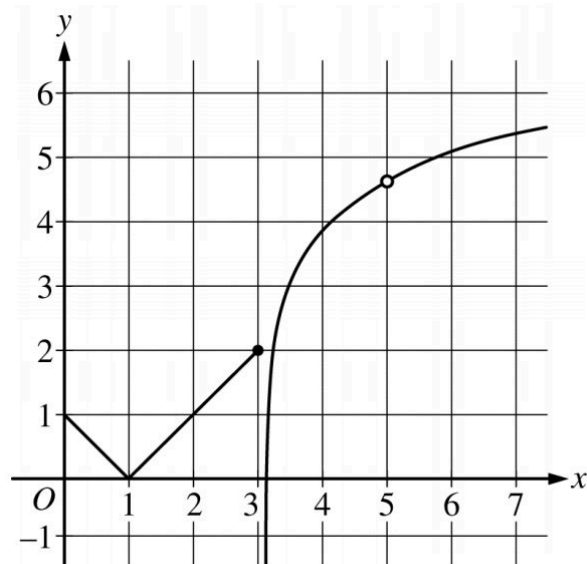
The graphs of the functions f and g are shown in the figures above.

Which of the following statements is false?

- (A) $\lim_{x \rightarrow 1} f(x) = 0$
- (B) $\lim_{x \rightarrow 2} g(x)$ does not exist.
- (C) $\lim_{x \rightarrow 1} (f(x)g(x+1))$ does not exist. ✓
- (D) $\lim_{x \rightarrow 1} (f(x+1)g(x))$ exists.

Unit 1 AP Topics 1-4

7.

Graph of f

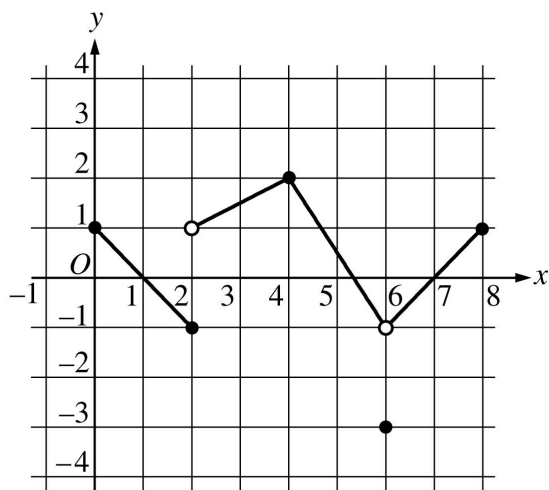
The graph of a function f is shown above. Which of the following limits does not exist?

- (A) $\lim_{x \rightarrow 1^-} f(x)$
- (B) $\lim_{x \rightarrow 1} f(x)$
- (C) $\lim_{x \rightarrow 3^-} f(x)$
- (D) $\lim_{x \rightarrow 3} f(x)$
- (E) $\lim_{x \rightarrow 5} f(x)$



Unit 1 AP Topics 1-4

8.



The figure above shows the graph of the function f . Which of the following statements are true?

I. $\lim_{x \rightarrow 2^-} f(x) = f(2)$

II. $\lim_{x \rightarrow 6^-} f(x) = \lim_{x \rightarrow 6^+} f(x)$

III. $\lim_{x \rightarrow 6} f(x) = f(6)$

(A) II only

(B) III only

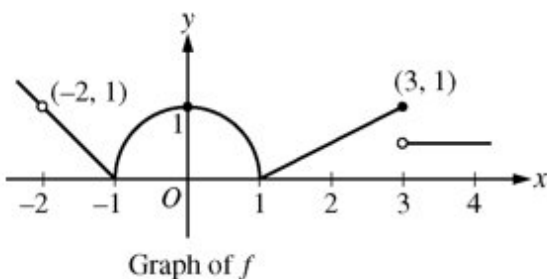
(C) I and II only

(D) II and III only

(E) I, II, and III



9.



The graph of a function f is shown above. For which of the following values of c does $\lim_{x \rightarrow c} f(x) = 1$?

(A) 0 only

(B) 0 and 3 only

(C) -2 and 0 only

(D) -2 and 3 only

(E) -2, 0, and 3



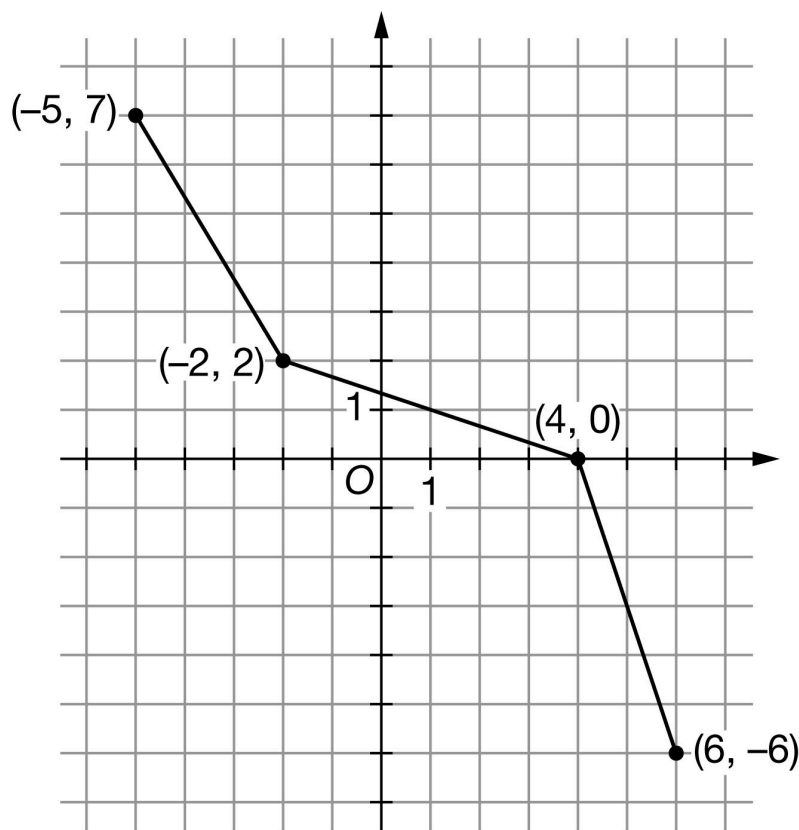
Unit 1 AP Topics 1-4

10. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the function f on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_{-2}^x f(t) \, dt$ for $-5 \leq x \leq 6$.

- For the function h , is $x = 4$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-5 < x < 6$ is the graph of h both increasing and concave down? Give a reason for your answer.
- Find the value of $h''(-2)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow -2} \frac{h(x)}{x-2}$. Show the computations that lead to your answer.

Unit 1 AP Topics 1-4

(f) Let k be the function defined by $k(x) = h(x^2)$. Find the value of $k'(\sqrt{5})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{5}$ can be keyboarded as sqrt(5))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local maximum or relative maximum. We also need to see a connection to h . The obvious is $h' = f$. A response of local maximum with the explanation: " f (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " h' changes from positive to negative," without the connection of $h' = f$ will not earn the point.

Simply saying " h . changes from increasing to decreasing" is not enough to earn the point.

In this situation, appealing to the Second Derivative Test by claiming that $h''(4) < 0$ is an incorrect statement, so a response with this reason does not earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the location of a relative maximum of h because $h'(x) = f(x)$ changes from positive to negative at this point.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 4$ is the only critical point. (f is positive on $(-5, 4)$ and negative on $(4, 6)$ is another way to state this.)

Note: Stating that $x = 4$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 4$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 4$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

Unit 1 AP Topics 1-4

The third point is only earned if the first two points are earned and the student presents the correct value of the maximum.

The presentation of a coordinate point $(4, 6)$ as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 4$. This consideration is more than just the statement $x = 4$. The response must use $x = 4$ and an area calculation in trying to find $h(4)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find areas for $h(-5)$ and $h(6)$.

The response cannot simply write the integrals with the correct upper limits. Attempts at both areas must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

Common errors that show an attempt to find area:

· $h(6) = 12$ (adding the triangle areas)

· $Ah(-5) = -\frac{15}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = -5$ by stating $h(-5)$ is negative will earn the second point as long as $x = 6$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = 4$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point $(4, 6)$ is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

☐ Identifies unique critical point

Unit 1 AP Topics 1-4

- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = 4$ is the only point on the interval $-5 \leq x \leq 6$ where $h'(x) = f(x) = 0$, $x = 4$ is the only critical point of h on the interval $-5 \leq x \leq 6$.

Because $x = 4$ is also the location of a relative maximum of h , $h(4)$ is the absolute maximum for h on $-5 \leq x \leq 6$.

$$h(4) = \int_{-2}^4 f(t) dt = 6$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 6$ is $h(4) = 6$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 4$ is the only critical point in the interval because it is the only point where $h'(x) = f(x) = 0$.

$$h(4) = \int_{-2}^4 f(t) dt = 6$$

$$h(-5) = \int_{-2}^{-5} f(t) dt = -\int_{-5}^{-2} f(t) dt = -13.5$$

$$h(6) = \int_{-2}^6 f(t) dt = 0$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 6$ is $h(4) = 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(-5, 4)$ or inequality $-5 < x < 4$.

We will allow $[-5, 4]$, $(-5, 4]$, $[-5, 4)$, or an equivalent inequality.

Presenting $(-5, -2) \cup (-2, 4)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The “ $\cup$ ” is not necessary. Be careful, one bracket on either -2 is

Unit 1 AP Topics 1-4

actually the entire interval and is eligible for both points.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons " h' is positive and decreasing" or " h' is positive and h'' is negative" will earn the second point if the connection $h' = f$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function h , then the second point is not earned.



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

The graph of h is both increasing and concave down on the open interval $(-5, 4)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $h''(x) = f'(x)$.

We are not allowing an import of a previous connection between h'' and f' .

The explicit connection, $h''(x) = f'(x)$, earns the first point.

It is not assumed that a response which only presents $h'(x) = f(x)$ has made the connection between h'' and f' .

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appear in these scoring guidelines,
- limits involving h' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of f at $x = -2$.

Unit 1 AP Topics 1-4

Note that all of the above descriptions earn the first point for the implicit consideration of $h''(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: f' does not exist because f (or "the graph") has a/is a (blank). Fill in the blank with corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of f . The phrase $h''(-2)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $h''(-2)$ does not exist because there is a cusp on f is acceptable and earns both points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(x)$
- ☐ Answer with explanation

Solution:

$$h''(x) = f'(x)$$

Because $\lim_{x \rightarrow -2^-} \frac{f(x) - f(-2)}{x - (-2)} = -\frac{5}{3}$ and $\lim_{x \rightarrow -2^+} \frac{f(x) - f(-2)}{x - (-2)} = -\frac{1}{3}$, $f'(-2) = h''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of h does not need to be stated.

An answer of $\frac{h(-2)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Unit 1 AP Topics 1-4

Disregard limit notation errors.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

h is continuous. $\Rightarrow \lim_{x \rightarrow -2} h(x) = h(-2)$

$$\lim_{x \rightarrow -2} \frac{h(x)}{x-2} = \frac{h(-2)}{-2-2} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : h'(x^2) \cdot 2x$.

This point is earned for any of $h'(x^2) \cdot 2x$, $f(x^2) \cdot 2x$, $f(5) \cdot 2\sqrt{5}$, or $-3 \cdot 2\sqrt{5}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but could still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$h'(x^2) = f(x^2), h'(5) = f(5), \text{ or } h'(5) = -3.$$

This point would be earned with a response of $f(5) \cdot 2\sqrt{5}$ or $-3 \cdot 2\sqrt{5}$ even with no preceding work.

The third point is for either $-3 \cdot 2\sqrt{5}$ or $-6\sqrt{5}$. Note, the product need not be simplified.

An answer of $-6\sqrt{5}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus

Unit 1 AP Topics 1-4☐ Answer**Solution:**

$$k'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$$

$$k'(\sqrt{5}) = f(5) \cdot 2\sqrt{5} = -3 \cdot 2\sqrt{5} = -6\sqrt{5}$$

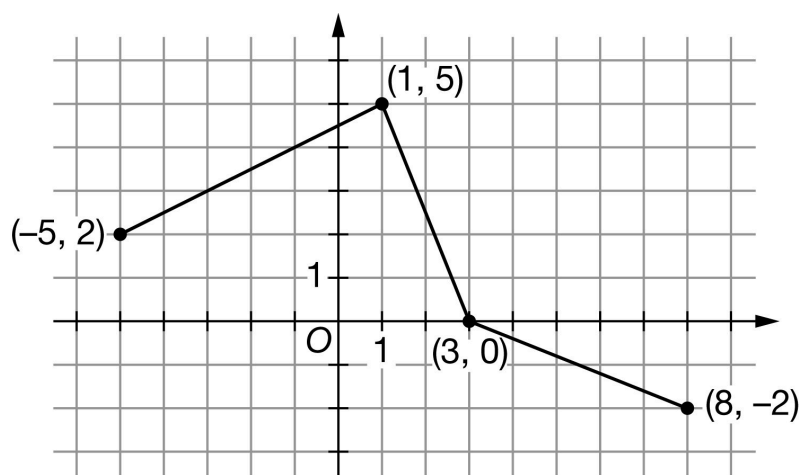
Unit 1 AP Topics 1-4

11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g on the interval $-5 \leq x \leq 8$ consists of three line segments, as shown in the figure above. Let m be the function defined by $m(x) = \int_1^x g(t) \, dt$ for $-5 \leq x \leq 8$.

- For the function m , is $x = 3$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute maximum value of m on the interval $-5 \leq x \leq 8$. Justify your answer.
- On what open intervals contained in $-5 < x < 8$ is the graph of m both increasing and concave down? Give a reason for your answer.
- Find the value of $m''(1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 1} \frac{m(x)}{x+1}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = m(x^2)$. Find the value of $k'(1)$ or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local maximum or relative maximum. We also need to see a connection to g . The obvious is $m' = g$. A response of local maximum with the explanation: " g (or the graph) changes from positive

Unit 1 AP Topics 1-4

to negative" is an implied connection. A response of local maximum with explanation: " m' changes from positive to negative" without the connection of $m' = g$ is not acceptable.

Simply saying " m changes from increasing to decreasing" is not acceptable.

In this situation, the Second Derivative Test $m''(3) < 0$ is an incorrect statement and the response with this reason does not earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a relative maximum of m because $m'(x) = g(x)$ changes from positive to negative there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 3$ is the only critical point. (g is positive on $(-5, 3)$ and negative on $(3, 8)$ is another way to state this.)

Note: Stating that $x = 3$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 3$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 3$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 3$. This consideration is more than just the statement $x = 3$. The response must use $x = 3$ and an area calculation in trying to find $m(3)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Unit 1 AP Topics 1-4

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $g(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

· $m(-5) = -9$ (finding the area of the triangle and forgetting the rectangle)

· $m(8) = 10$ (adding the triangle areas). These can be used to earn the second point.

A correct argument that eliminates $x = -5$ by stating $m(-5)$ is negative will earn the second point as long as $x = 8$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = 3$, and $x = 8$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

NOTE: A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x = 3$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = 3$ is the only point on the interval $-5 \leq x \leq 8$ where $m'(x) = g(x) = 0$, $x = 3$ is the only critical point of m on the interval $-5 \leq x \leq 8$.

Unit 1 AP Topics 1-4

Because $x = 3$ is also the location of a relative maximum for m , $m(3)$ is the absolute maximum for m in $-5 \leq x \leq 8$.

$$m(3) = \int_1^3 g(t) dt = 5$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 3$ is the only critical point in the interval because it is the only point where $m'(x) = g(x) = 0$.

$$m(3) = \int_1^3 g(t) dt = 5$$

$$m(-5) = \int_1^{-5} g(t) dt = - \int_{-5}^1 g(t) dt = -21$$

$$m(8) = \int_1^8 g(t) dt = 0$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(1, 3)$ or inequality $1 < x < 3$.

We will allow $[1, 3]$, $(1, 3]$, $[1, 3)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons m' is positive and decreasing or m' is positive and m'' is negative, will earn the second point if the connection $m' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function m , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

Reason

Unit 1 AP Topics 1-4

**Solution:**

m is increasing where $m' = g$ is positive.

m is concave down where $m' = g$ is decreasing.

The graph of m is increasing and concave down on the open interval $(1, 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $m''(x) = g'(x)$.

We are not allowing an import of a previous connection between m'' and g' .

The explicit connection, $m''(x) = g'(x)$, earns the first point.

It is not assumed that a response which only presents $m'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g' , the differentiability of g , or the characteristic of the graph of g .

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving g' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of g at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g . The phrase $m''(1)$ of 1 does not exist because there is a cusp will not earn this point. Whereas, the phrase $m''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.

Unit 1 AP Topics 1-4



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

$$m''(x) = g'(x)$$

Because $\lim_{x \rightarrow 1^-} \frac{g(x) - g(1)}{x - 1} = \frac{1}{2}$ and $\lim_{x \rightarrow 1^+} \frac{g(x) - g(1)}{x - 1} = -\frac{5}{2}$, $g'(1) = m''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of m does not need to be stated.

An answer of $\frac{m(1)}{2}$ will not earn the point,

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{2}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$m \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 1} m(x) = m(1)$$

Unit 1 AP Topics 1-4

$$\lim_{x \rightarrow 1} \frac{m(x)}{x+1} = \frac{m(1)}{1+1} = \frac{0}{2} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : m'(x^2) \cdot 2x$.

This point is earned for any of $m'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(1) \cdot 2$, or $5 \cdot 2$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$m'(x^2) = g(x^2), m'(1) = g(1), \text{ or } m'(1) = 5.$$

This point would be earned with a response of $g(1) \cdot 2$, or $5 \cdot 2$, even with no preceding work.

The third point is for either $5 \cdot 2$ or 10. Note, the product need not be simplified.

An answer of 10 with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = m'(x^2) \cdot 2x = g(x^2) \cdot 2x$$

$$k'(1) = g(1) \cdot 2 = 5 \cdot 2 = 10$$

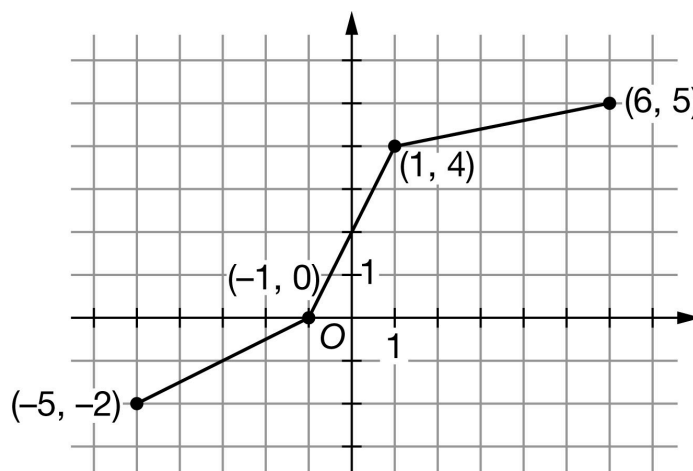
Unit 1 AP Topics 1-4

12. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the function h on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let p be the function defined by $p(x) = \int_1^x h(t) \, dt$ for $-5 \leq x \leq 6$.

- For the function p , is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of p on the interval $-5 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-5 < x < 6$ is the graph of p both decreasing and concave up? Give a reason for your answer.
- Find the value of $p''(1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 1} \frac{p(x)}{x+4}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = p(x^2)$. Find the value of $k'(\sqrt{2})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{2}$ can be keyboarded as sqrt(2))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to h . The obvious is $p' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative

Unit 1 AP Topics 1-4

to positive" is an implied connection. A response of local minimum with explanation: " p' changes from negative to positive," without the connection of $p' = h$ is not acceptable.

Simply saying " p changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $p''(-1) > 0$ is an incorrect statement and a response with this reason would not earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of p because $p'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (h is negative on $(-5, -1)$ and positive on $(-1, 6)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that they do not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ and an area calculation in trying to find $p(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Unit 1 AP Topics 1-4

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $h(x)$ values.

A common error that shows an attempt to find area is $k(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

· $p(-5) = 8$ (adding the triangle areas)

· $p(6) = \frac{5}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $p(6)$ is positive will earn the second point as long as $x = -5$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = -1$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -1$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Unit 1 AP Topics 1-4

Because $x = -1$ is the only point on the interval $-5 \leq x \leq 6$ where $p'(x) = h(x) = 0$, $x = -1$ is the only critical point of p on the interval $-5 \leq x \leq 6$.

Because $x = -1$ is also the location of a relative minimum for p , $p(-1)$ is the absolute minimum for p on $-5 \leq x \leq 6$.

$$p(-1) = \int_1^{-1} h(t) dt = - \int_{-1}^1 h(t) dt = -4$$

Therefore, the absolute minimum value of p on $-3 \leq x \leq 6$ is $p(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $p'(x) = h(x) = 0$.

$$p(-1) = \int_1^{-1} h(t) dt = - \int_{-1}^1 h(t) dt = -4$$

$$p(-5) = \int_1^{-5} h(t) dt = - \int_{-5}^1 h(t) dt = 0$$

$$p(6) = \int_1^6 h(t) dt = 22.5$$

Therefore, the absolute minimum value of p on $-3 \leq x \leq 6$ is $p(-1) = -4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-5, -1)$ or inequality $-5 < x < -1$.

We will allow $[-5, -1]$, $(-5, -1]$, $[-5, -1)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons p' is negative and increasing or p' is negative and p'' is positive will earn the second point if the connection $p' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function p , then the second point is not earned.



0	1	2
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The student response accurately includes **both** of the criteria below.

Unit 1 AP Topics 1-4

- ☐ Answer
- ☐ Reason

Solution:

p is decreasing where $p' = h$ is negative.

p is concave up where $p' = h$ is increasing.

The graph of p is both decreasing and concave up on the open interval $(-5, -1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $p''(x) = h'(x)$.

We are not allowing an import of a previous connection between p'' and h' .

The explicit connection, $p'' = h'$, earns the first point.

It is not assumed that a response which only presents $p' = h$ has made that connection.

The implicit connection will be made in the explanation with a connection to h' , the differentiability of h , or the characteristic of the graph of h .

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving h' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of h at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $p''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $p''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.

Unit 1 AP Topics 1-4



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

$$p''(x) = h'(x)$$

Because $\lim_{x \rightarrow 1^-} \frac{h(x) - h(1)}{x - 1} = 2$, and $\lim_{x \rightarrow 1^+} \frac{h(x) - h(1)}{x - 1} = \frac{1}{5}$, $h'(1) = p''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of p does not need to be stated.

An answer of $\frac{p(1)}{5}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{5}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors



0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$p \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 1^-} p(x) = p(1) \lim_{x \rightarrow 1} \frac{p(x)}{x+4} = \frac{p(1)}{1+4} = \frac{0}{5} = 0$$

Unit 1 AP Topics 1-4

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of k : $p'(x^2) \cdot 2x$.

This point is earned for any of $p'(x^2) \cdot 2x$, $h(x^2) \cdot 2x$, $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$p'(x^2) = h(x^2), p'(2) = h(2), \text{ or } p'(2) = \frac{21}{5}.$$

This point would be earned with a response of $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, even with no preceding work.

The third point is for either $\frac{21}{5} \cdot 2\sqrt{2}$ or $\frac{42}{5}\sqrt{2}$. Note, the product need not be simplified.

An answer of $\frac{42}{5}\sqrt{2}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = p'(x^2) \cdot 2x = h(x^2) \cdot 2x$$

$$k'(\sqrt{2}) = h(2) \cdot 2\sqrt{2} = \frac{21}{5} \cdot 2\sqrt{2} = \left(\frac{42}{5}\right)\sqrt{2}$$

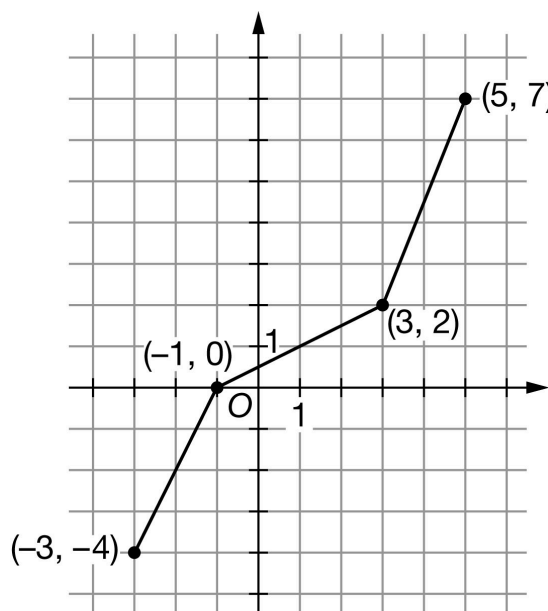
Unit 1 AP Topics 1-4

13. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g on the interval $-3 \leq x \leq 5$ consists of three line segments, as shown in the figure above. Let k be the function defined by $k(x) = \int_3^x g(t) \, dt$ for $-3 \leq x \leq 5$.

- For the function k , is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of k on the interval $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals contained in $-3 < x < 5$ is the graph of k both increasing and concave up? Give a reason for your answer.
- Find the value of $k''(-1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 3} \frac{k(x)}{x+3}$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = k(x^2)$. Find the value of $h'(\sqrt{3})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as sqrt(3))

Part A

Unit 1 AP Topics 1-4

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to g . The obvious is $k' = g$. A response of local minimum with the explanation: “ g (or the graph) changes from negative to positive” is an implied connection. A response of local minimum with explanation: “ k' changes from negative to positive,” without the connection of $k' = g$ is not acceptable.

Simply saying “ k changes from decreasing to increasing” is not acceptable.

In this situation, the Second Derivative Test $k''(-1) > 0$ is an incorrect statement and a response with this reason does not earn the point.



0	1
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The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of k because $k'(x) = g(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (g is negative on $(-3, -1)$ and positive on $(-1, 5)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

Unit 1 AP Topics 1-4

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ AND an area calculation in trying to find $k(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

A common error that shows an attempt to find area is $k(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words “must check the endpoints” is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

- $k(-3) = 8$ (adding the triangle areas)
- $k(5) = 5$ (finding the area of the triangle and forgetting the rectangle)
- These can be used to earn the second point.

A correct argument that eliminates $x = 5$ by stating $k(5)$ is positive will earn the second point as long as $x = -3$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -3$, $x = -1$, and $x = 5$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -1$

Unit 1 AP Topics 1-4

- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = -1$ is the only point on the interval $-3 \leq x \leq 5$ where $k'(x) = g(x) = 0$, $x = -1$ is the only critical point of k on the interval $-3 \leq x \leq 5$.

Because $x = -1$ is also the location of a relative minimum for k , $k(-1)$ is the absolute minimum for k on $-3 \leq x \leq 5$.

$$k(-1) = \int_3^{-1} g(t)dt = -\int_{-1}^3 g(t)dt = -4$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $k'(x) = g(x) = 0$.

$$k(-1) = \int_3^{-1} g(t)dt = -\int_{-1}^3 g(t)dt = -4$$

$$k(-3) = \int_3^{-3} g(t)dt = -\int_{-3}^3 g(t)dt = 0$$

$$k(5) = \int_3^5 g(t)dt = 9$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-1, 5)$ or inequality $-1 < x < 5$.

We will allow $[-1, 5]$, $(-1, 5]$, $[-1, 5)$, or an equivalent inequality.

Presenting $(-1, 3) \cup (3, 5)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The “ $\cup$ ” is not necessary. Be careful, one bracket on either 3 is actually the entire interval and is eligible for both points.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

Unit 1 AP Topics 1-4

The response which reasons “ k' is positive and increasing” or “ k' is positive and k'' is positive” will earn the second point if the connection $k' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function k , then the second point is not earned.

The interval $(-1, 7)$ with the correct justification will earn one of the two points. (The response took the y -coordinate of the right-hand point of the interval $(5, 7)$). This is considered a copy error. This is the only copy error that is allowed.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

k is increasing where $k' = g$ is positive.

k is concave up where $k' = g$ is increasing.

The graph of k is both increasing and concave up on the open interval $(-1, 5)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $k''(x) = g'(x)$.

We are not allowing an import of a previous connection between k'' and g' .

The explicit connection, $k''(x) = g'(x)$ earns the first point.

It is not assumed that a response which only presents $k'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g' , the differentiability of g , or the characteristic of the graph of g .

The second point is for the answer with explanation.

This must include “Does not exist” and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving g' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of g at $x = -1$.

Unit 1 AP Topics 1-4

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or “the graph”) has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will NOT accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g . The phrase $k''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $k''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

Because $\lim_{x \rightarrow -1^-} \frac{g(x) - g(-1)}{x - (-1)} = 2$ and $\lim_{x \rightarrow -1^+} \frac{g(x) - g(-1)}{x - (-1)} = \frac{1}{2}$, $g'(-1) = k''(-1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of k does not need to be stated.

An answer of $\frac{k(3)}{6}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{6}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors.

Unit 1 AP Topics 1-4



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

k is continuous. $\Rightarrow \lim_{x \rightarrow 3} k(x) = k(3)$

$$\lim_{x \rightarrow 3} \frac{k(x)}{x+3} = \frac{k(3)}{3+3} = \frac{0}{6} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $h : k'(x^2) \cdot 2x$.

This point is earned for any of $k'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$k'(x^2) = g(x^2), k'(3) = g(3), \text{ or } k'(3) = 2.$$

This point would be earned with a response of $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $2 \cdot 2\sqrt{3}$, or $4\sqrt{3}$. Note, the product need not be simplified.

An answer of $4\sqrt{3}$ with no supporting work earns no points in part (f).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Unit 1 AP Topics 1-4**Solution:**

$$h'(x) = k'(x^2) \cdot 2x = g(x^2) \cdot 2x$$

$$h'(\sqrt{3}) = g(3) \cdot 2\sqrt{3} = 2 \cdot 2\sqrt{3} = 4\sqrt{3}$$

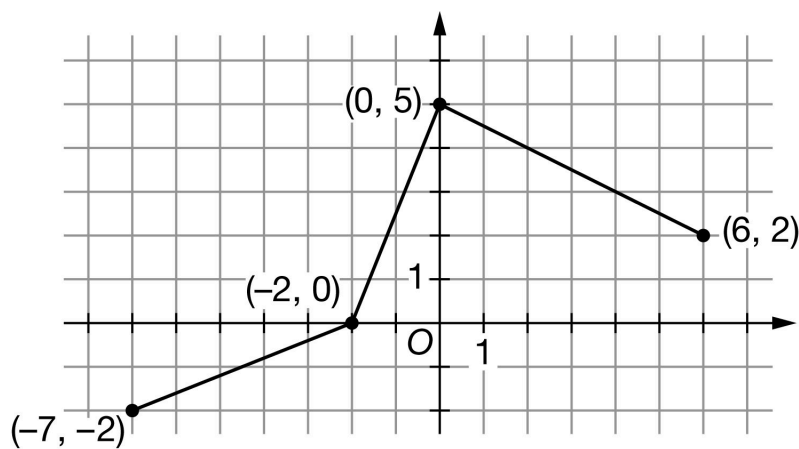
Unit 1 AP Topics 1-4

14. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the function h on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let r be the function defined by $r(x) = \int_0^x h(t) \, dt$ for $-7 \leq x \leq 6$.

- For the function r , is $x = -2$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of r on the interval $-7 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-7 < x < 6$ is the graph of r both increasing and concave down? Give a reason for your answer.
- Find the value of $r''(-2)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 0} \frac{r(x)}{x-4}$. Show the computations that lead to your answer.
- Let v be the function defined by $v(x) = r(x^2)$. Find the value of $v'(\sqrt{3})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as sqrt(3))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to h . The obvious is $r' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative

Unit 1 AP Topics 1-4

to positive" is an implied connection. A response of local minimum with explanation: " r' changes from negative to positive," without the connection of $r' = h$ is not acceptable.

Simply saying " r changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $r''(2) > 0$ is an incorrect statement, and a response with this reason does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -2$ is the location of a relative minimum of r because $r'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -2$ is the only critical point. (h is negative on $(-7, -2)$ and positive on $(-2, 6)$ is another way to state this.)

Note: Stating that $x = -2$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -2$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -2$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -2$. This consideration is more than just the statement $x = -2$. The response must use $x = -2$ and an area calculation in trying to find $r(-2)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Unit 1 AP Topics 1-4

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $h(x)$ values.

A common error that shows an attempt to find area is $r(-2) = 5$. This earns the first point.

The second point is for consideration of the endpoints along with an attempt to find area. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

· $r(-7) = 10$ (adding the triangle areas)

· $r(6) = 9$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $r(6)$ is positive will earn the second point as long as $x = -7$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -7$, $x = -2$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

Note: A coordinate point is not an acceptable answer.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Unit 1 AP Topics 1-4**Solution:**

Because $x = -2$ is the only point on the interval $-7 \leq x \leq 6$ where $r'(x) = h(x) = 0$, $x = -2$ is the only critical point of r on the interval $-7 \leq x \leq 6$.

Because $x = -2$ is also the location of a relative minimum for r , $r(-2)$ is the absolute minimum for r on $-7 \leq x \leq 6$.

$$r(-2) = \int_0^{-2} h(t) dt = - \int_{-2}^0 h(t) dt = -5$$

Therefore, the absolute minimum value of r on $-7 \leq x \leq 6$ is $r(-2) = -5$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -2$ is the only critical point in the interval because it is the only point where $r'(x) = h(x) = 0$.

$$r(-2) = \int_0^{-2} h(t) dt = - \int_{-2}^0 h(t) dt = -5$$

$$r(-7) = \int_0^{-7} h(t) dt = - \int_{-7}^0 h(t) dt = 0$$

$$r(6) = \int_0^6 h(t) dt = 21$$

Therefore, the absolute minimum value of r on $-7 \leq x \leq 6$ is $r(-2) = -5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(0, 6)$ or inequality $0 < x < 6$.

We will allow $[0, 6]$, $(0, 6]$, $[0, 6)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons r' is positive and decreasing or r' is positive and r'' is negative will earn the second point if the connection $r' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function r , then the second point is not earned.

Unit 1 AP Topics 1-4



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

r is increasing where $r' = h$ is positive.

r is concave down where $r' = h$ is decreasing.

The graph of r is both increasing and concave down on the open interval $(0, 6)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a consideration of $r''(x) = h'(x)$.

We are not allowing an import of a previous connection between r'' and h' .

The explicit connection $r''(x) = h'(x)$ earns the first point.

It is not assumed that a response which presents $r'(x) = h(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to h' , the differentiability of h , or the characteristic of the graph of h .

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving h' as the argument of the limit, or some correct verbal description of the characteristic of the graph of h at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a

Unit 1 AP Topics 1-4

node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $r''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $r''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

Because $\lim_{x \rightarrow -2^-} \frac{h(x) - h(-2)}{x - (-2)} = \frac{2}{5}$ and $\lim_{x \rightarrow -2^+} \frac{h(x) - h(-2)}{x - (-2)} = \frac{5}{2}$, $h'(-2) = r''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of r does not need to be stated.

An answer of $\frac{r(0)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors



0	1
---	---

The student response accurately includes the criteria below.

Unit 1 AP Topics 1-4

☐ Answer
Solution:
 r is continuous. $\Rightarrow \lim_{x \rightarrow 0} r(x) = r(0)$

$$\lim_{x \rightarrow 0} \frac{r(x)}{x-4} = \frac{r(0)}{0-4} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $v : r'(x^2) \cdot 2x$.

This point is earned for any of $r'(x^2) \cdot 2x$, $h(x^2) \cdot 2x$, $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$r'(x^2) = h(x^2), r'(3) = h(3), \text{ or } r'(3) = \frac{7}{2}.$$

This point would be earned with a response of $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $\frac{7}{2} \cdot 2\sqrt{3}$ or $7\sqrt{3}$. Note, the product need not be simplified.

An answer of $7\sqrt{3}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
☐ Fundamental Theorem of Calculus
☐ Answer

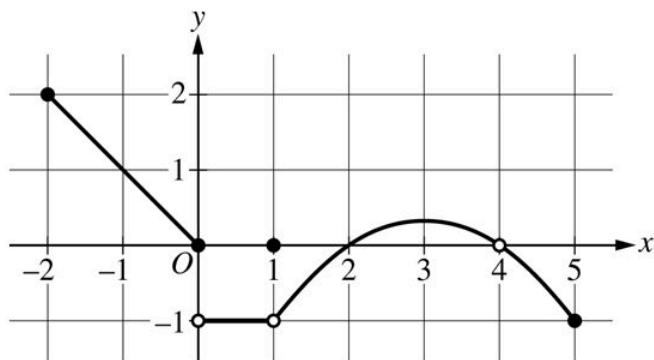
Solution:

$$v'(x) = r'(x^2) \cdot 2x = h(x^2) \cdot 2x$$

$$v'(\sqrt{3}) = h(3) \cdot 2\sqrt{3} = \frac{7}{2} \cdot 2\sqrt{3} = 7\sqrt{3}$$

Unit 1 AP Topics 1-4

15.

Graph of f

The graph of the function f is shown above. For what values of a does $\lim_{x \rightarrow a} f(x) = 0$?

- (A) 2 only
(B) 2 and 4
(C) 0 and 2 only
(D) 0, 1, and 2

**Answer B**

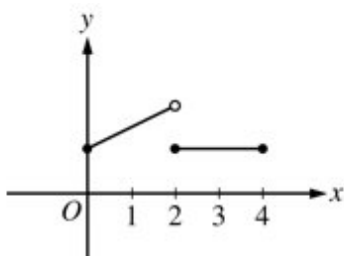
This option is correct. $\lim_{x \rightarrow a} f(x) = 0$ if the left-sided and right-sided limits

$$x = a \text{ at } x = a \text{ both equal } 0 \text{ i.e. } \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = 0$$

For this graph, that happens at $x = 2$ and $x = 4$. Note that the function does not need to be defined at $x = 4$ for the limit to exist there.

Unit 1 AP Topics 1-4

16.

Graph of f

The figure above shows the graph of a function f with domain $0 \leq x \leq 4$. Which of the following statements are true?

I. $\lim_{x \rightarrow 2^-} f(x)$ exists.

II. $\lim_{x \rightarrow 2^+} f(x)$ exists.

III. $\lim_{x \rightarrow 2} f(x)$ exists.

(A) I only

(B) II only

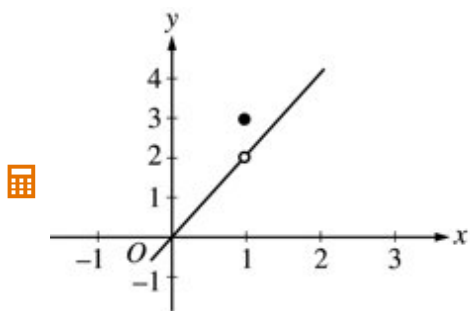
(C) I and II only

(D) I and III only

(E) I, II, and III



17.

Graph of f

The graph of the function f is shown in the figure above. The value of $\lim_{x \rightarrow 1} \sin(f(x))$ is

(A) 0.909

(B) 0.841

(C) 0.141

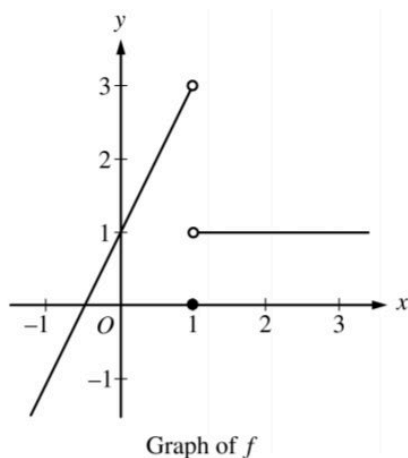
(D) -0.416

(E) nonexistent



Unit 1 AP Topics 1-4

18.



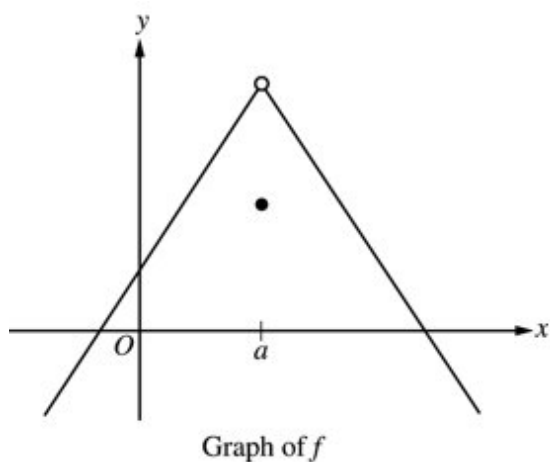
The graph of $y = f(x)$ is shown above.

What is $\lim_{x \rightarrow 1} f(x)$?

- (A) 0
- (B) 1
- (C) 3
- (D) The limit does not exist.



19.



The graph of the function f is shown above. Which of the following statements must be false?

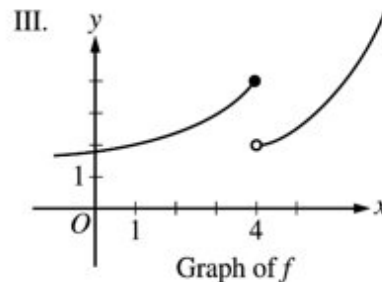
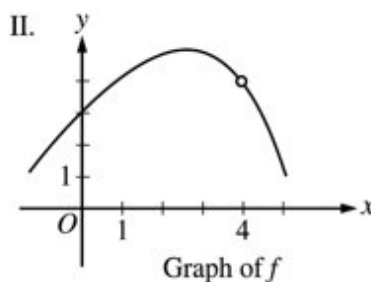
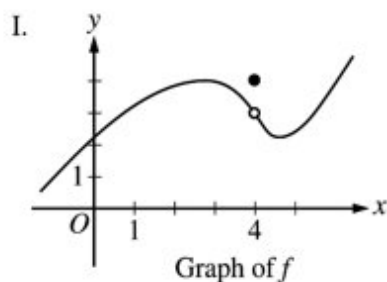
- (A) $f(a)$ exists.
- (B) $f(x)$ is defined for $0 < x < a$.
- (C) f is not continuous at $x = a$.
- (D) $\lim_{x \rightarrow a} f(x)$ exists.

- (E) $\lim_{x \rightarrow a} f'(x)$ exists.



Unit 1 AP Topics 1-4

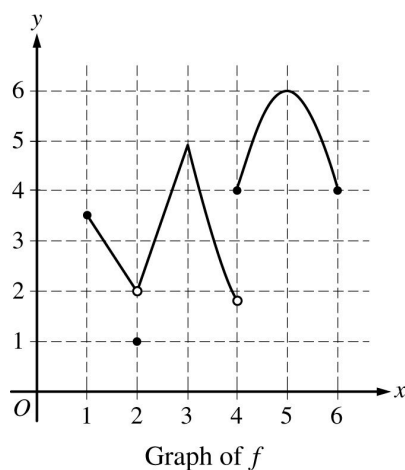
20. For which of the following does $\lim_{x \rightarrow 4} f(x)$ exist?



- (A) I only
 (B) II only
 (C) III only
 (D) I and II only
 (E) I and III only



21.



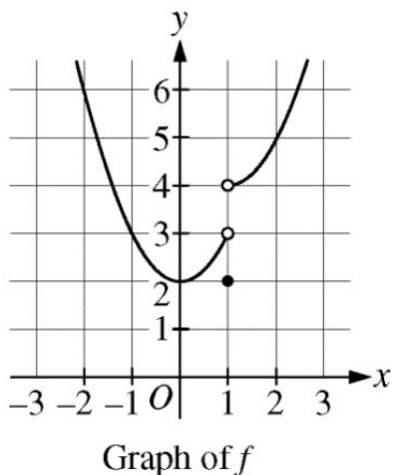
The graph of the function f is shown above. Which of the following statements is false?

- (A) $\lim_{x \rightarrow 2} f(x)$ exists.
 (B) $\lim_{x \rightarrow 3} f(x)$ exists.
 (C) $\lim_{x \rightarrow 4} f(x)$ exists.
 (D) $\lim_{x \rightarrow 5} f(x)$ exists.
 (E) The function f is continuous at $x = 3$.



Unit 1 AP Topics 1-4

22.



The graph of the function f is shown in the figure above. The value of $\lim_{x \rightarrow 0} f(1 - x^2)$ is

- (A) 1
 (B) 2
 (C) 3
 (D) 4
 (E) nonexistent



23.

x	-0.1	-0.01	-0.001	0.001	0.01	0.1
$f(x)$	-0.1054	-0.0101	-0.001	0.001	0.0099	0.0953

The function f is continuous and increasing for $x \geq -1$. The table above gives values of f at selected values of x .

Of the following, which is the best approximation for $\lim_{x \rightarrow 0} e^{-2f(x)}$?

- (A) -2
 (B) 0
 (C) 1
 (D) The limit does not exist



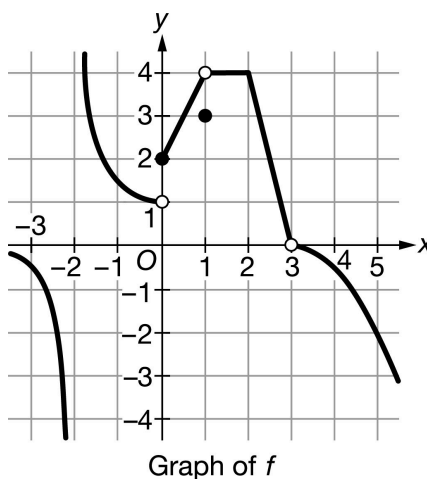
Unit 1 AP Topics 1-4

24. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



x	0.8	0.9	0.99	0.999	1.001	1.01	1.1	1.2
$g(x)$	4.16	4.59	4.960	4.996	5.004	5.040	5.39	5.76

The graph of the function f is shown in the xy -plane above. The graph of f has a vertical asymptote at $x = -2$.

The function g is continuous and increasing for all x . Values of $g(x)$ at selected values of x are shown in the table above.

- Using the graph of f and the table for g , estimate $\lim_{x \rightarrow 1} (2f(x) + 3g(x))$.
- For each of the values $a = -2$, $a = 2$, and $a = 3$, determine whether or not f is continuous at $x = a$. In each case, use the definition of continuity to justify your answer.
- Find the value of $\lim_{x \rightarrow 0} f(f(x))$ or explain why the limit does not exist.
- Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 1$.

Part A

Both limits must be correct to earn the answer point. The response may still earn the points with use of equals rather than approximately. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 1-4



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ estimate for $\lim_{x \rightarrow 1} g(x)$
- ☐ answer

Solution:

$$\lim_{x \rightarrow 1} f(x) = 4$$

$$\lim_{x \rightarrow 1} g(x) \approx 5$$

$$\lim_{x \rightarrow 1} (2f(x) + 3g(x)) = 2\lim_{x \rightarrow 1} f(x) + 3\lim_{x \rightarrow 1} g(x) \approx 2 \cdot 4 + 3 \cdot 5 = 23$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ not continuous at $x = -2$ with reason
- ☐ continuous at $x = 2$ with reason
- ☐ not continuous at $x = 3$ with reason

Solution:

$$\lim_{x \rightarrow -2} f(x) \text{ does not exist because } \lim_{x \rightarrow -2^-} f(x) = -\infty \text{ and } \lim_{x \rightarrow -2^+} f(x) = \infty.$$

Therefore, f is not continuous at $x = -2$

$$\lim_{x \rightarrow 2} f(x) = 4 = f(2)$$

Therefore, f is continuous at $x = 2$

f is not defined at $x = 3$ Therefore, f is not continuous at $x = 3$.

Unit 1 AP Topics 1-4

Part C

At most 1 out of 3 points if mathematical notation for limit is missing or incorrect. Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ considers one-sided limits
- ☐ answer

Solution:

As x approaches 0 from the left, $f(x)$ approaches 1 from above. Therefore, the limit of $f(x)$ should be evaluated as x approaches 1 from the right, as shown.

$$\lim_{x \rightarrow 0^-} f(f(x)) = \lim_{x \rightarrow 1^+} f(x) = 4$$

As x approaches 0 from the right, $f(x)$ approaches 2 from above. Therefore, the limit of $f(x)$ should be evaluated as x approaches 2 from the right, as shown.

$$\lim_{x \rightarrow 0^+} f(f(x)) = \lim_{x \rightarrow 2^+} f(x) = 4$$

Therefore, $\lim_{x \rightarrow 0} f(f(x)) = 4$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(1.001) - g(0.999)}{1.001 - 0.999} = \frac{5.004 - 4.996}{1.001 - 0.999}$$

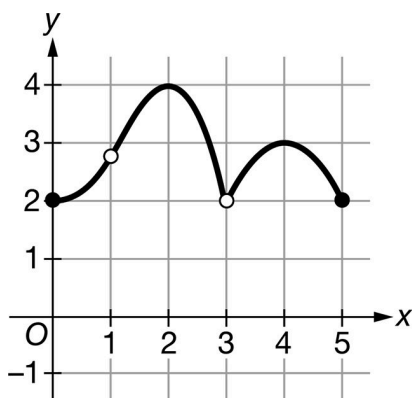
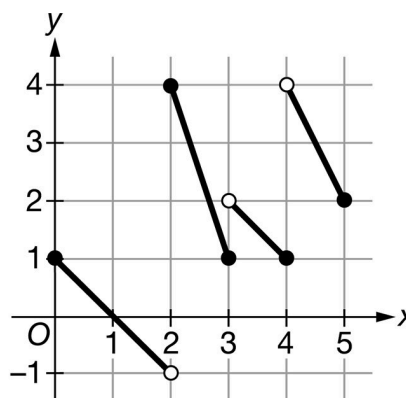
Unit 1 AP Topics 1-4

25. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f Graph of g

The graphs of the functions f and g are shown above on the interval $0 \leq x \leq 5$.

- (a) Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 2.5$.
- (b) Let h be the function defined by $h(x) = g(f(x))$. Find the value of $\lim_{x \rightarrow 3} h(x)$ or explain why the limit does not exist. Use correct limit notation in your answer.
- (c) Let k be the function defined by $k(x) = (4 - f(x))g(x)$. Consider $x = 2$ and $x = 4$. Determine whether k is continuous at each of these values. Justify your answers using correct limit notation.
- (d) Values of $f(x)$ at selected values of x are given in the table below. What type of discontinuity does f have at $x = 1$? Based on the values in the table, what is a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$? Give a reason for your answer.

x	0.9	0.99	0.999	1	1.001	1.01	1.1
$f(x)$	2.634	2.770	2.784	Undefined	2.787	2.801	2.946

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0

1

Unit 1 AP Topics 1-4

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(3)-g(2)}{3-2} = \frac{1-4}{3-2}$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ considers one-sided limits
- ☐ answer

Solution:

As x approaches 3 from the left and from the right, $f(x)$ approaches 2 from above. Therefore, the limit of $g(x)$ should be evaluated as x approaches 2 from the right, as shown.

$$\lim_{x \rightarrow 3^-} h(x) = \lim_{x \rightarrow 3^-} g(f(x)) = \lim_{x \rightarrow 2^+} g(x) = 4$$

$$\lim_{x \rightarrow 3^+} h(x) = \lim_{x \rightarrow 3^+} g(f(x)) = \lim_{x \rightarrow 2^+} g(x) = 4$$

Therefore, $\lim_{x \rightarrow 3} h(x) = 4$.

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes two of the criteria below.

Unit 1 AP Topics 1-4

- ☐ continuous at $x = 2$ with justification
- ☐ not continuous at $x = 4$ with justification

Solution:

$$\lim_{x \rightarrow 2^-} k(x) = \lim_{x \rightarrow 2^-} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 2^-} f(x)\right) \lim_{x \rightarrow 2^-} g(x) = (4 - 4)(-1) = 0$$

$$\lim_{x \rightarrow 2^+} k(x) = \lim_{x \rightarrow 2^+} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 2^+} f(x)\right) \lim_{x \rightarrow 2^+} g(x) = (4 - 4)(4) = 0$$

$$k(2) = (4 - f(2))g(2) = (4 - 4)(4) = 0$$

Because $\lim_{x \rightarrow 2} k(x) = 0 = k(2)$, k is continuous at $x = 2$.

$$\lim_{x \rightarrow 4^-} k(x) = \lim_{x \rightarrow 4^-} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 4^-} f(x)\right) \lim_{x \rightarrow 4^-} g(x) = (4 - 3)(1) = 1$$

$$\lim_{x \rightarrow 4^+} k(x) = \lim_{x \rightarrow 4^+} (4 - f(x))g(x) = \left(4 - \lim_{x \rightarrow 4^+} f(x)\right) \lim_{x \rightarrow 4^+} g(x) = (4 - 3)(4) = 4$$

Because $\lim_{x \rightarrow 4^-} k(x) \neq \lim_{x \rightarrow 4^+} k(x)$,

$\lim_{x \rightarrow 4} k(x)$ does not exist and therefore k is not continuous at $x = 4$.

Part D

Any value between 2.784 and 2.787 earns the estimate point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ identifies removable discontinuity
- ☐ estimate
- ☐ reason

Solution:

The graph f of indicates that f has a removable discontinuity at $x = 1$. The graph shows that f is increasing near $x = 1$, so $2.784 < \lim_{x \rightarrow 1} f(x) < 2.787$. A reasonable estimate for $\lim_{x \rightarrow 1} f(x)$ would be determined by averaging the

Unit 1 AP Topics 1-4

function values on each side of 1.

The estimate is as follows.

$$\lim_{x \rightarrow 1} f(x) \approx \frac{f(0.999) + f(1.001)}{2} = \frac{2.784 + 2.787}{2} = 2.7855$$

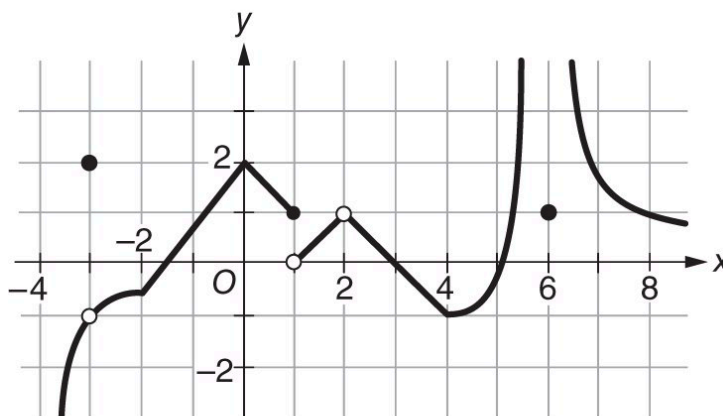
Unit 1 AP Topics 1-4

26. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

x	-0.2	-0.1	-0.01	-0.001	0.001	0.01	0.1	0.2
$g(x)$	1.975	1.988	1.997	1.999	2.001	2.003	2.012	2.025

The graph of the function f is shown in the xy -plane above. f has a discontinuity due to a vertical asymptote at $x = 6$.

Let g be a function that is continuous and increasing for all x . Values of $g(x)$ at selected values of x are shown in the table above.

- Using the graph of f and the table for g , estimate $\lim_{x \rightarrow 0} (f(x) - g(x))$.
- For each of the values $a = 1$, $a = 2$, $a = 4$, and $a = 6$, find $\lim_{x \rightarrow a} f(x)$ or explain why $\lim_{x \rightarrow a} f(x)$ does not exist. Use correct limit notation in your answers.
- Is f continuous at $x = -3$? Using correct notation, justify your answer.
- Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 0$.

Part A

Both limits must be correct to earn the answer point. The response may still earn the points with use of equals rather than approximately.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 1-4



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ estimate for $\lim_{x \rightarrow 0} g(x)$
- ☐ answer

Solution:

$$\lim_{x \rightarrow 0} f(x) = 2 \text{ and } \lim_{x \rightarrow 0} g(x) \approx 2$$

$$\lim_{x \rightarrow 0} (f(x) - g(x)) = \lim_{x \rightarrow 0} f(x) - \lim_{x \rightarrow 0} g(x) \approx 2 - 2 = 0$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ does not exist at $a = 1$ with reason
- ☐ values at $a = 2$ and $a = 4$
- ☐ does not exist at $a = 6$ with reason

Solution:

$$\lim_{x \rightarrow 1^-} f(x) = 1 \text{ and } \lim_{x \rightarrow 1^+} f(x) = 0$$

$$\lim_{x \rightarrow 1} f(x) \text{ does not exist because}$$

$$\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x).$$

$$\lim_{x \rightarrow 2} f(x) = 1$$

$$\lim_{x \rightarrow 4} f(x) = -1$$

Unit 1 AP Topics 1-4

$\lim_{x \rightarrow 6} f(x)$ does not exist because

$$\lim_{x \rightarrow 6^-} f(x) = \infty \text{ and } \lim_{x \rightarrow 6^+} f(x) = \infty.$$

Part C

At most 1 out of 3 points if mathematical notation for limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ limit value
- ☐ answer with justification

Solution:

$$\lim_{x \rightarrow -3} f(x) = -1 \text{ and } f(-3) = 2$$

No, f is not continuous at $x = -3$ because $\lim_{x \rightarrow -3} f(x) \neq f(-3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(0.001) - g(-0.001)}{0.001 - (-0.001)} = \frac{2.001 - 1.999}{0.001 - (-0.001)}$$

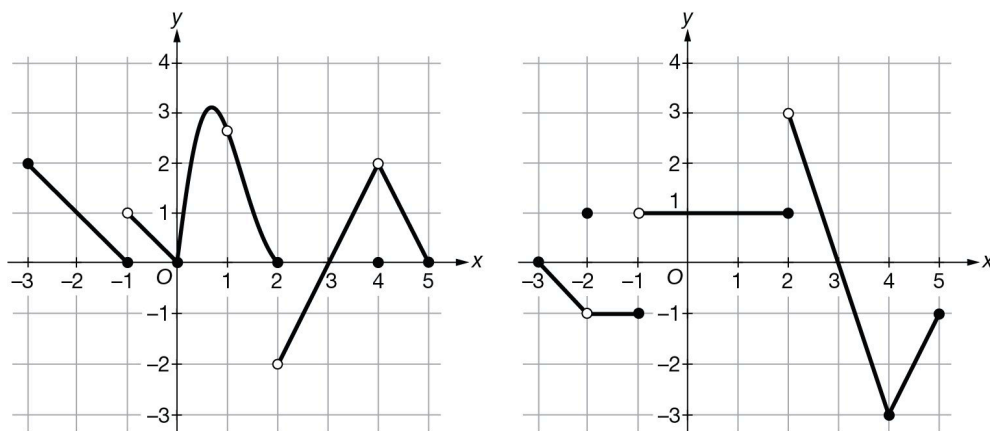
Unit 1 AP Topics 1-4

27. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graphs of the functions f and g are shown above on the interval $-3 \leq x \leq 5$. The graphs consist of line segments, except where $f(x) = \frac{5 \sin(\pi x)}{2 - 2x^3}$ on the interval $0 \leq x \leq 2$.

- Write a difference quotient that best approximates the instantaneous rate of change of g at $x = 4.5$.
- Let $h(x) = g(f(x))$. Find $\lim_{x \rightarrow 4} h(x)$. Use correct limit notation in your answer.
- Let $k(x) = f(x) + g(x)$. Consider $x = -1$, $x = 2$, and $x = 4$. At which one of these values does k have a jump discontinuity? Justify your answer for the jump discontinuity using correct limit notation.
- Values of $f(x)$ at selected values of x are given in the table below. What type of discontinuity does f have at $x = 1$? Based on the values in the table, what is a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$? Give a reason for your answer.

x	0.995	0.996	0.997	0.998	0.999	1	1.001	1.002	1.003	1.004	1.005
$f(x)$	2.631	2.628	2.626	2.623	2.621	Undefined	2.615	2.613	2.610	2.607	2.605

Part A

Select a point value to view scoring criteria, solutions, **and/or** examples to score the response.

Unit 1 AP Topics 1-4



0	1
---	---

The student response accurately includes a correct difference quotient.

Solution:

$$\frac{g(5)-g(4)}{5-4} = \frac{(-1)-(-3)}{5-4}$$

Part B

At most 2 out of 3 points if mathematical notation for any limit is missing or incorrect.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ limit notation
- ☐ considers one-sided limits
- ☐ answer

Solution:

As x approaches 4 from the left and from the right, $f(x)$ approaches 2 from below. Therefore, the limit of $g(x)$ should be evaluated as x approaches 2 from the left, as shown.

$$\lim_{x \rightarrow 4^-} h(x) = \lim_{x \rightarrow 4^-} g(f(x)) = \lim_{x \rightarrow 2^-} g(x) = 1$$

$$\lim_{x \rightarrow 4^+} h(x) = \lim_{x \rightarrow 4^+} g(f(x)) = \lim_{x \rightarrow 2^-} g(x) = 1$$

Therefore, $\lim_{x \rightarrow 4} h(x) = 1$.

Part C

At most 1 out of 2 points if mathematical notation for any limit is missing or incorrect OR if response includes jump discontinuity at maximum of one incorrect value.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Unit 1 AP Topics 1-4



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ jump discontinuity at $x = -1$
- ☐ justification, using correct limit notation

Solution:

There is a jump discontinuity at $x = -1$ because both one-sided limits exist at $x = -1$ and are finite but are not equal:

$$\lim_{x \rightarrow -1^-} k(x) = \lim_{x \rightarrow -1^-} f(x) + \lim_{x \rightarrow -1^-} g(x) = 0 + (-1) = -1$$

$$\lim_{x \rightarrow -1^+} k(x) = \lim_{x \rightarrow -1^+} f(x) + \lim_{x \rightarrow -1^+} g(x) = 1 + 1 = 2$$

Note: k does not have a jump discontinuity at $x = 2$. $\lim_{x \rightarrow 2^-} k(x) = 0 + 1 = 1$ and

$$\lim_{x \rightarrow 2^+} k(x) = -2 + 3 = 1$$

Note: k does not have a jump discontinuity at $x = 4$. $\lim_{x \rightarrow 4^-} k(x) = 2 + (-3) = -1$

and $\lim_{x \rightarrow 4^+} k(x) = 2 + (-3) = -1$

Part D

Any value between 2.615 and 2.621 earns the estimate point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ identifies removable discontinuity
- ☐ estimate
- ☐ reason

Unit 1 AP Topics 1-4**Solution:**

The graph of f indicates that f has a removable discontinuity at $x = 1$.

The graph shows that f is decreasing near $x = 1$, so $2.621 > \lim_{x \rightarrow 1} f(x) > 2.615$. Since the table of values reflects approximate linear behavior, a reasonable estimate for $\lim_{x \rightarrow 1} f(x)$ would be determined by averaging the function values on each side of 1.

The estimate is as follows. $\lim_{x \rightarrow 1} f(x) \approx \frac{f(0.999) + f(1.001)}{2} = \frac{2.621 + 2.615}{2} = 2.618$

Alternately,

a reasonable estimate can be found by observing that as x increases by 0.001, $f(x)$ decreases by 0.002 or 0.003. Since 2.618 is 0.003 from both 2.621 and 2.615, it is the best estimate.

Unit 1 AP Topics 1-4

28. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

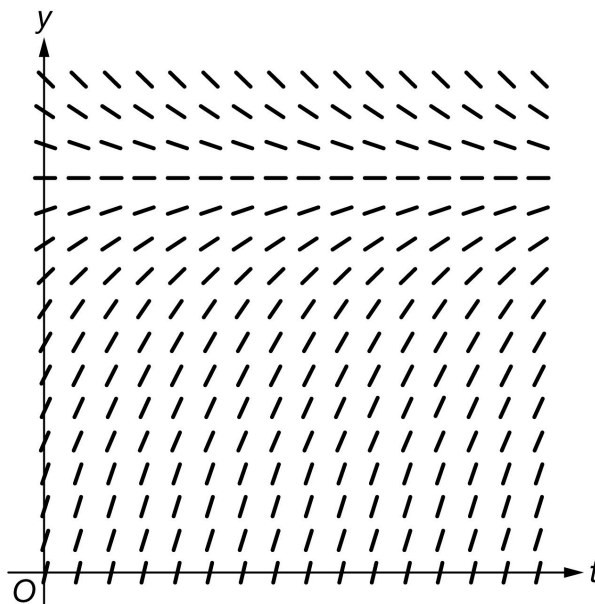
Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

A medication is administered to a patient. The amount, in milligrams, of the medication in the patient at time t hours is modeled by a function $y = A(t)$ that satisfies the differential equation $\frac{dy}{dt} = \frac{12-y}{3}$. At time $t = 0$ hours, there are 0 milligrams of the medication in the patient.

- (a) A portion of the slope field for the differential equation $\frac{dy}{dt} = \frac{12-y}{3}$ is given below. Sketch the solution curve through the point $(0, 0)$.



- (b) Using correct units, interpret the statement $\lim_{t \rightarrow \infty} A(t) = 12$ in the context of this problem.
- (c) Use separation of variables to find $y = A(t)$, the particular solution to the differential equation $\frac{dy}{dt} = \frac{12-y}{3}$ with initial condition $A(0) = 0$.
- (d) A different procedure is used to administer the medication to a second patient. The amount, in milligrams, of the medication in the second patient at time t hours is modeled by a function $y = B(t)$ that satisfies the differential equation $\frac{dy}{dt} = 3 - \frac{y}{t+2}$. At time $t = 1$ hour, there are 2.5 milligrams of the medication in the second patient. Is the rate of change of the amount of medication in the second patient increasing or decreasing at time $t = 1$? Give a reason for your answer.

Part A

Unit 1 AP Topics 1-4

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the solution curve must pass through the point $(0, 0)$, be generally increasing and concave down, and approach the horizontal asymptote from below as t increases. The point is not earned if two or more solution curves are presented.

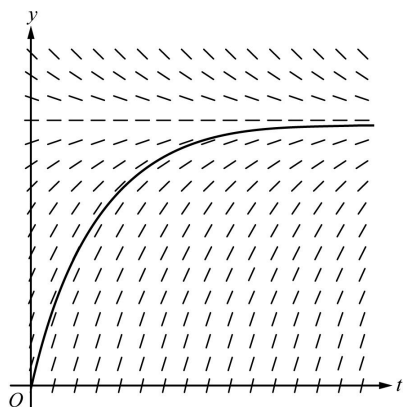


0	1
---	---

The student response accurately includes the criteria below.

☐ Solution curve

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the point the interpretation must include “medication in the patient,” “approaches 12,” and units (milligrams), or their equivalents.



0	1
---	---

The student response accurately includes the criteria below.

☐ Interpretation

Solution:

Over time the amount of medication in the patient approaches 12 milligrams.

Unit 1 AP Topics 1-4

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{dy}{12-y} = 3 dt$ is a bad separation and does not earn the first point. However, this response is eligible for the second and third points. It cannot earn the fourth point.

Absolute value bars are not required in this part.

A response that correctly separates to $\frac{3 dy}{12-y} = dt$ but then incorrectly simplifies to $\frac{dy}{4-y} = dt$ earns the first point (for the initial correct separation), is eligible for the second point (for $-\ln|4-y| = t$, with or without $+C$), but is not eligible for the third or fourth points.

$+\ln|12-y| = \frac{t}{3}$ (with or without $+C$) does not earn the second point and is not eligible for the fourth point;
 $+\ln|12-y| = \frac{t}{3} + C$ is eligible for the third point

In all other cases, the points are earned consecutively—the second point cannot be earned without the first, the third without the second, etc.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

- ☐ Separation of variables
- ☐ Antiderivatives
- ☐ Constant of integration and uses initial condition
- ☐ Solves for y

Solution:

$$\frac{dy}{dt} = \frac{12-y}{3} \Rightarrow \frac{dy}{12-y} = \frac{dt}{3}$$

$$\int \frac{dy}{12-y} = \int \frac{dt}{3} \Rightarrow -\ln|12-y| = \frac{t}{3} + C$$

$$\ln|12-y| = -\frac{t}{3} - C \Rightarrow |12-y| = e^{-t/3-C}$$

$$\Rightarrow y = 12 + Ke^{-t/3}$$

$$0 = 12 + K \Rightarrow K = -12$$

$$y = A(t) = 12 - 12e^{-t/3}$$

Unit 1 AP Topics 1-4

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is for correctly applying the quotient rule to $\frac{y}{t+2}$ or applying the product rule to $y(t+2)^{-1}$. Errors in differentiating the constant, 3, or handling the sign of the second term of $\frac{dy}{dt}$ will result in not earning the second point.

The second point cannot be earned unless the second derivative $\frac{d^2y}{dt^2}$ is correct.

For the second point it is sufficient to state the sign of $B''(1)$ is negative with supporting work. If a value is declared for $B''(1)$, it must be correct in order to earn the second point.

Eligibility for the third point: An attempt at using the quotient rule (or product rule) to find $B''(1)$. In this case the third point will be earned for a consistent conclusion based on the declared value (or sign) of $B''(1)$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Quotient rule
- ☐ $B''(1) < 0$
- ☐ Answer with reason

Solution:

$$\frac{dy}{dt} = 3 - \frac{y}{t+2} \Rightarrow \frac{d^2y}{dt^2} = (-1) \frac{\frac{dy}{dt}(t+2) - y}{(t+2)^2}$$

$$B'(1) = 3 - \frac{B(1)}{3} = 3 - \frac{2.5}{3} = \frac{6.5}{3}$$

$$B''(1) = -\frac{B'(1) \cdot 3 - B(1)}{3^2} = -\frac{6.5 - 2.5}{9} = -\frac{4}{9} < 0$$

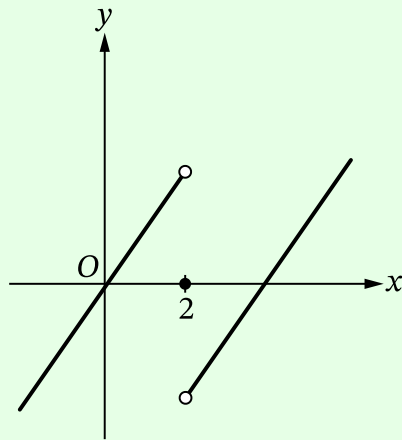
The rate of change of the amount of medication is decreasing at time $t = 1$ because $B''(1) < 0$ and $\frac{d^2y}{dt^2}$ is continuous in an interval containing $t = 1$.

Unit 1 AP Topics 1-4

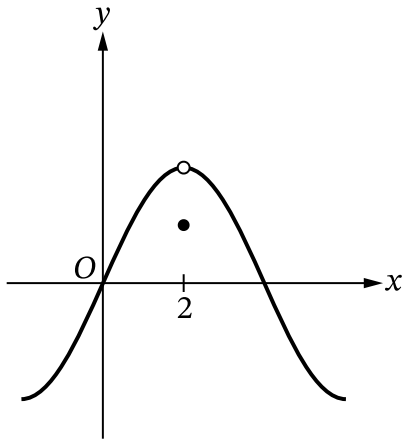
29. Which of the following is the graph of a function f such that $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$ both exist, but $\lim_{x \rightarrow 2} f(x)$ does not exist?

Unit 1 AP Topics 1-4

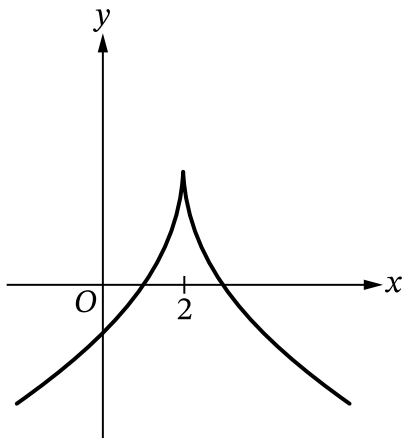
(A)



(B)

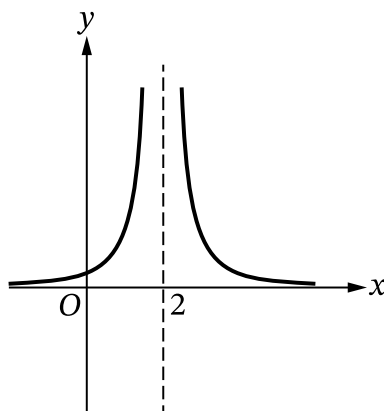


(C)



Unit 1 AP Topics 1-4

(D)

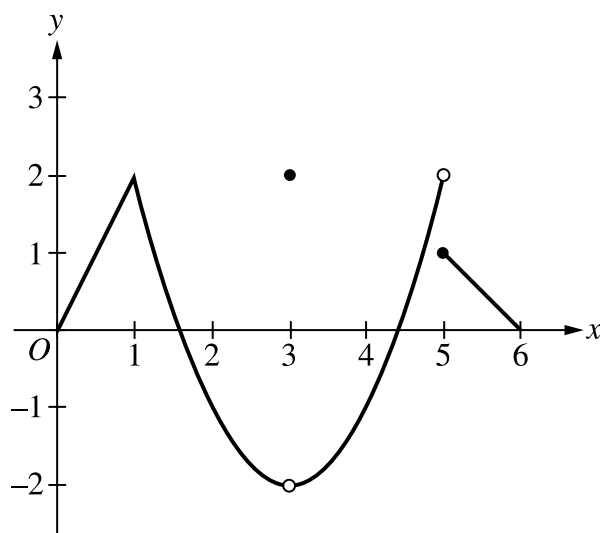
**Answer A**

Correct. For this graph, $\lim_{x \rightarrow 2^-} f(x)$ exists and is a positive value, and $\lim_{x \rightarrow 2^+} f(x)$ exists but is a negative value.

Therefore $\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$, so $\lim_{x \rightarrow 2} f(x)$ does not exist.

Unit 1 AP Topics 1-4

30.

Graph of f

The graph of the function f is shown. For which of the following values of a does $\lim_{x \rightarrow a} f(x)$ not exist?

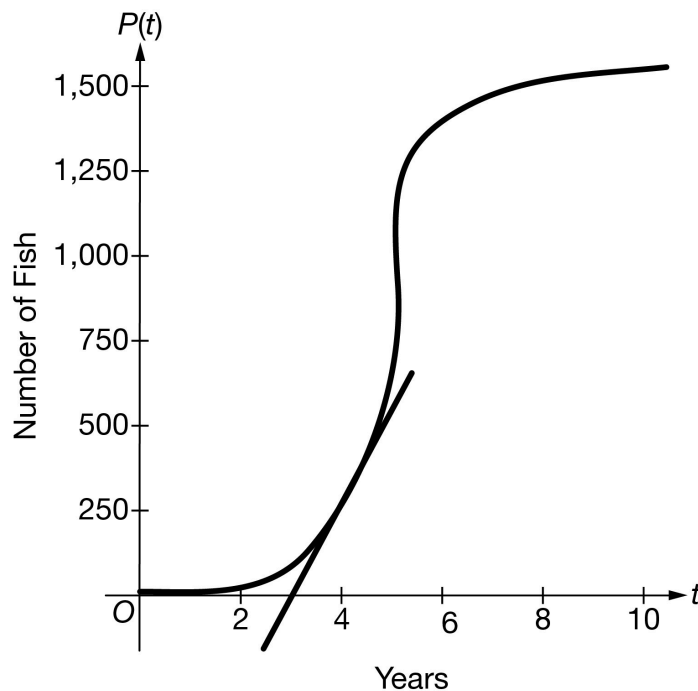
- (A) 1 only
(B) 5 only
(C) 3 and 5 only
(D) 1, 3, and 5

**Answer B**

Correct. $\lim_{x \rightarrow 5} f(x)$ does not exist because $\lim_{x \rightarrow 5^-} f(x) = 2$ and $\lim_{x \rightarrow 5^+} f(x) = 1$, so $\lim_{x \rightarrow 5^-} f(x) \neq \lim_{x \rightarrow 5^+} f(x)$.

Unit 1 AP Topics 1-4

31.



The size of a population of fish in a pond is modeled by the function P , where $P(t)$ gives the number of fish and t gives the number of years after the first year of introduction of the fish to the pond for $0 \leq t \leq 10$. The graph of the function P and the line tangent to P at $t = 4$ are shown above. Which of the following gives the best estimate for the instantaneous rate of change of P at $t = 4$?

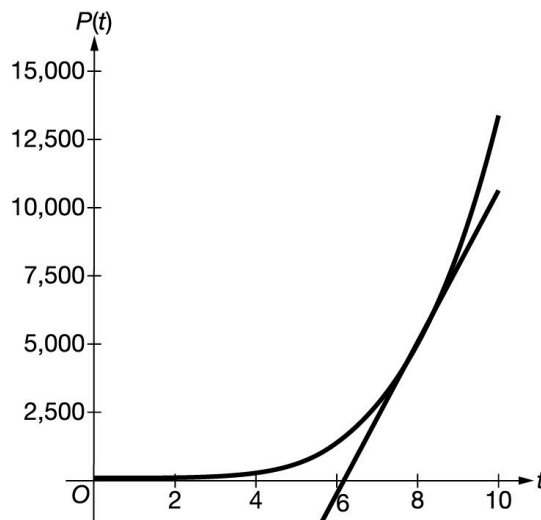
- (A) $P(4)$
- (B) The slope of the line joining $(0, P(0))$ and $(4, P(4))$
- (C) The slope of the line joining $(0, P(0))$ and $(8, P(8))$
- (D) The slope of the line joining $(3.9, P(3.9))$ and $(4.1, P(4.1))$

**Answer D**

Correct. If $h > 0$ is close to 0, we expect the slope of the line joining $(4 - h, P(4 - h))$ and $(4 + h, P(4 + h))$ to be close to the slope of the tangent line at $t = 4$.

Unit 1 AP Topics 1-4

32.



The size of a population of bacteria is modeled by the function P , where $P(t)$ gives the number of bacteria and t gives the number of hours after midnight for $0 \leq t \leq 10$. The graph of the function P and the line tangent to P at $t = 8$ are shown above. Which of the following gives the best estimate for the instantaneous rate of change of P at $t = 8$?

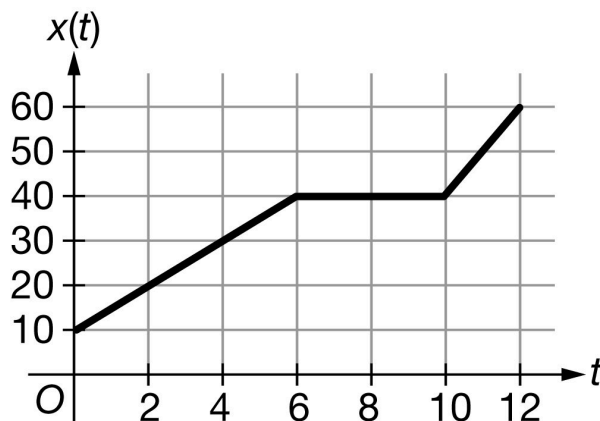
- (A) $P(8)$
- (B) The slope of the line joining $(0, P(0))$ and $(8, P(8))$
- (C) The slope of the line joining $(0, P(0))$ and $(10, P(10))$
- (D) The slope of the line joining $(7.9, P(7.9))$ and $(8.1, P(8.1))$

**Answer D**

Correct. If $h > 0$ is close to 0, we expect the slope of the line joining $(8 - h, P(8 - h))$ and $(8 + h, P(8 + h))$ to be close to the slope of the tangent line at $t = 8$.

Unit 1 AP Topics 1-4

33.



A particle is moving on the x -axis and the position of the particle at time t is given by $x(t)$, whose graph is shown above. Which of the following is the best estimate for the speed of the particle at time $t = 4$?

(A) 0

(B) 5

(C) $\frac{15}{2}$

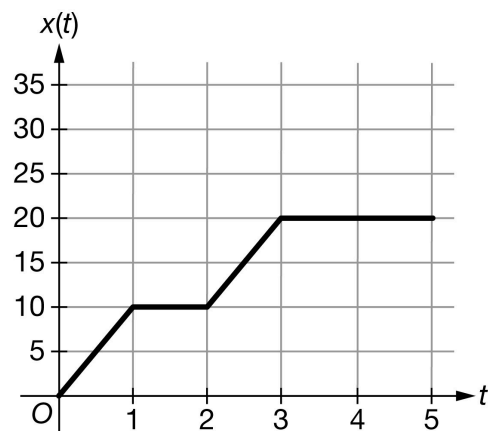
(D) 10

**Answer B**

Correct. Since the position function is linear, the slope of the graph at time t in the interval $[0, 6]$ is the speed of the particle at time $t = 4$.

Unit 1 AP Topics 1-4

34.



A particle is moving on the x -axis and the position of the particle at time t is given by $x(t)$, whose graph is given above. Which of the following is the best estimate for the speed of the particle at time $t = 4$?

(A) 0

(B) 5

(C) $\frac{20}{3}$

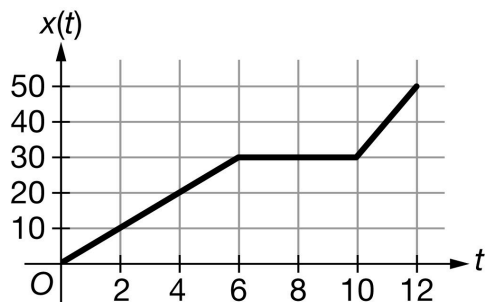
(D) 20

**Answer A**

Correct. At time $t = 4$, the particle is stationary at the position $x(t) = 20$.

Unit 1 AP Topics 1-4

35.



A particle is moving on the x -axis and the position of the particle at time t is given by $x(t)$, whose graph is given above. Which of the following is the best estimate for the speed of the particle at time $t = 8$?

(A) 0

(B) $\frac{15}{4}$

(C) 5

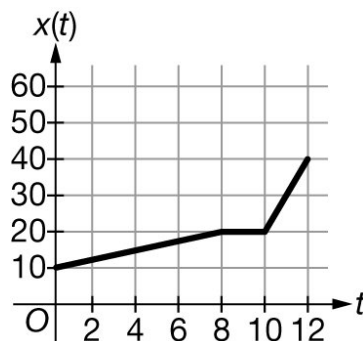
(D) 30

**Answer A**

Correct. At time $t = 8$, the particle is stationary at position $x(t) = 30$.

Unit 1 AP Topics 1-4

36.



A particle is moving on the x -axis, and the position of the particle at time t is given by $x(t)$, whose graph is shown above. Which of the following is the best estimate for the speed of the particle at time $t = 6$?

(A) 0

(B) $\frac{5}{4}$ (C) $\frac{35}{12}$ (D) $\frac{10}{3}$ **Answer B**

Correct. Since the position function is linear, the slope of the graph at time t in the interval $[0, 8]$ is the speed of the particle at time $t = 6$.

37. A car is driven on a straight road, and the distance traveled by the car after time $t = 0$ is given by a quadratic function s , where $s(t)$ is measured in feet and t is measured in seconds for $0 \leq t \leq 12$. Of the following, which gives the best estimate of the velocity of the car, in feet per second, at time $t = 6$ seconds?

(A) $s(6)$ (B) $\frac{s(6)}{6}$ (C) $\frac{s(8)-s(4)}{8-4}$ (D) $\frac{s(7)-s(5)}{7-5}$ **Answer D**

Correct. This gives the average velocity of the car over the shortest given time interval containing $t = 6$.

Unit 1 AP Topics 1-4

38. The position of a particle moving to the right on the x -axis is given by $x(t)$, where $x(t)$ is measured in centimeters and t is measured in seconds for $0 \leq t \leq 50$. If $y = x(t)$ is a linear function, which of the following would most likely give the best estimate of the speed of the particle, in centimeters per second, at time $t = 10$ seconds?
- (A) $x(10)$
- (B) $\frac{x(10)}{10}$
- (C) $x(11) - x(9)$
- (D) The slope of the graph of $y = x(t)$ ✓

Answer D

Correct. This gives the exact value of the speed of the particle at any time t in the interval $[0, 50]$.

39. An automobile is driven on a straight road, and the distance traveled by the automobile after time $t = 0$ is given by a quadratic function s , where $s(t)$ is measured in feet and t is measured in seconds for $0 \leq t \leq 12$. Of the following, which gives the best estimate of the velocity of the automobile, in feet per second, at time $t = 8$ seconds?
- (A) $s(8)$
- (B) $\frac{s(8)}{8}$
- (C) $\frac{s(12) - s(2)}{12 - 2}$
- (D) $\frac{s(9) - s(7)}{9 - 7}$ ✓

Answer D

Correct. This gives the average velocity of the automobile over the shortest given time interval containing $t = 8$.

Unit 1 AP Topics 1-4

40. The position of a particle moving to the right on the x -axis is given by $x(t)$, where $x(t)$ is measured in inches and t is measured in minutes for $0 \leq t \leq 100$. If $y = x(t)$ is a linear function, which of the following would most likely give the best estimate of the speed of the particle, in inches per minute, at time $t = 20$ minutes?
- (A) $x(20)$
- (B) $\frac{x(20)}{20}$
- (C) $x(21) - x(19)$
- (D) The slope of the graph of $y = x(t)$ ✓

Answer D

Correct. This gives the exact value of the speed of the particle at any time t in the interval $[0, 100]$.

41.

t (seconds)	0	100	200	300	400	500	600
$y(t)$ (feet)	0	50	400	1360	3200	6250	10,950

A rocket leaves the surface of Earth at time $t = 0$ and travels straight up from the surface. The height, in feet, of the rocket above the surface of Earth is given by $y(t)$, where t is measured in seconds for $0 \leq t \leq 600$. Values of $y(t)$ for selected values of t are given in the table above. Of the following values of t , at which value would the speed of the rocket most likely be greatest based on the data in the table?

- (A) $t = 100$
- (B) $t = 200$
- (C) $t = 300$
- (D) $t = 400$ ✓

Answer D

Correct. Considering the differences between successive values of $y(t)$ in the table, it appears that the speed of the rocket is increasing as t increases. $y(400) - y(300)$ gives the largest difference, considering the four data values.

Unit 1 AP Topics 1-4

42.

t (seconds)	0	20	40	60	80	100	120
$y(t)$ (feet)	0	105	300	900	2400	5000	10,000

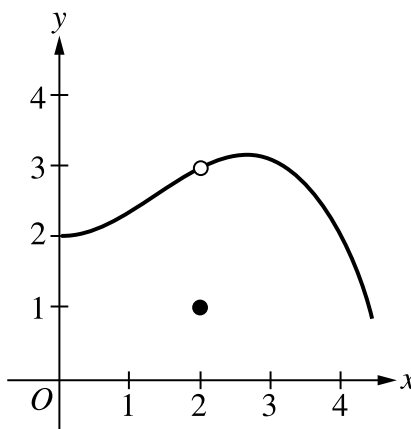
A model rocket leaves the ground at time $t = 0$ and travels straight up from the ground. The height, in feet, of the rocket above the ground is given by $y(t)$, where t is measured in seconds for $0 \leq t \leq 120$. Values of $y(t)$ for selected values of t are given in the table above. Of the following values of t , at which value would the velocity of the rocket most likely be greatest based on the data in the table?

(A) $t = 20$ (B) $t = 40$ (C) $t = 60$ (D) $t = 80$ **Answer D**

Correct. Considering the four values of t in the options and the differences between successive values of $y(t)$ in the table, note that the differences are increasing, with $y(80) - y(60)$ being the greatest difference. Therefore, the velocity of the rocket appears to be increasing as t increases, and so the greatest velocity would occur at the greatest of the four values of t , which is 80.

Unit 1 AP Topics 1-4

43.

Graph of f

The graph of the function f is shown. What is $\lim_{x \rightarrow 2} f(x)$?

(A) 0

(B) 1

(C) 3

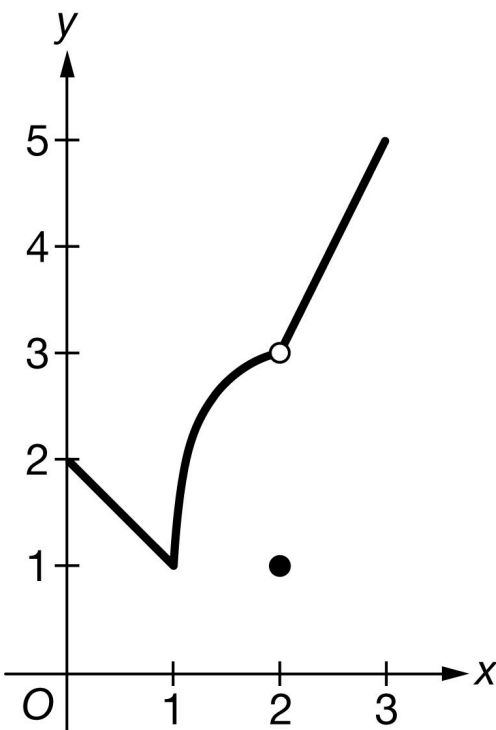
(D) The limit does not exist.

**Answer C**

Correct. The limit from the left is $\lim_{x \rightarrow 2^-} f(x) = 3$ and the limit from the right is $\lim_{x \rightarrow 2^+} f(x) = 3$. Because the one-sided limits are the same, $\lim_{x \rightarrow 2} f(x)$ exists and $\lim_{x \rightarrow 2} f(x) = 3$.

Unit 1 AP Topics 1-4

44.

Graph of f

The graph of the function f is shown above. What is $\lim_{x \rightarrow 2} f(x)$?

- (A) 1
(B) 2
(C) 3
(D) The limit does not exist.

**Answer C**

Correct. The limit at $x = 2$ will exist if the left- and right-sided limits are equal.

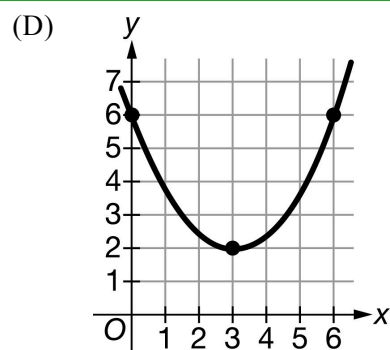
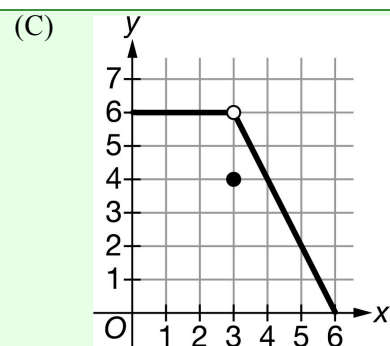
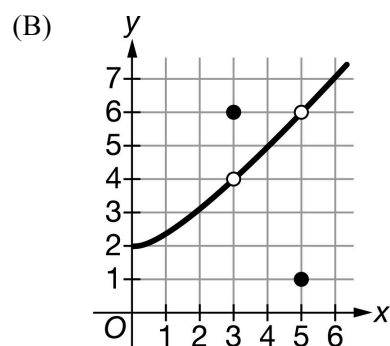
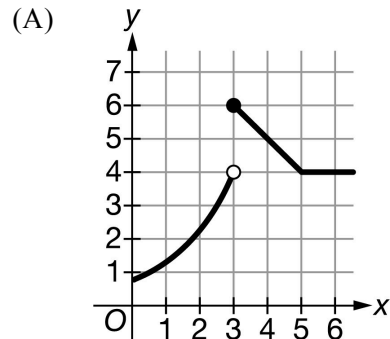
Left-sided limit: $\lim_{x \rightarrow 2^-} f(x) = 3$

Right-sided limit: $\lim_{x \rightarrow 2^+} f(x) = 3$

Since these limits are equal, $\lim_{x \rightarrow 2} f(x) = 3$.

Unit 1 AP Topics 1-4

45. A function f satisfies $\lim_{x \rightarrow 3} f(x) = 6$. Which of the following could be the graph of f ?



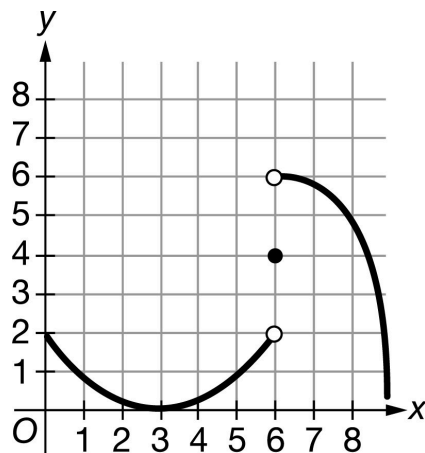
Answer C

Correct. In this graph, the values of $f(x)$ approach 6 as x approaches 3 from the left and as x approaches 3 from the right; that is, $\lim_{x \rightarrow 3^-} f(x) = 6 = \lim_{x \rightarrow 3^+} f(x)$. Since the one-sided limits have the common value 6, $\lim_{x \rightarrow 3} f(x)$ exists and

Unit 1 AP Topics 1-4

is equal to 6.

46.

Graph of f

The graph of the function f is shown above. Which of the following expressions equals 2 ?

(A) $f(6)$

(B) $\lim_{x \rightarrow 6^-} f(x)$

(C) $\lim_{x \rightarrow 6^+} f(x)$

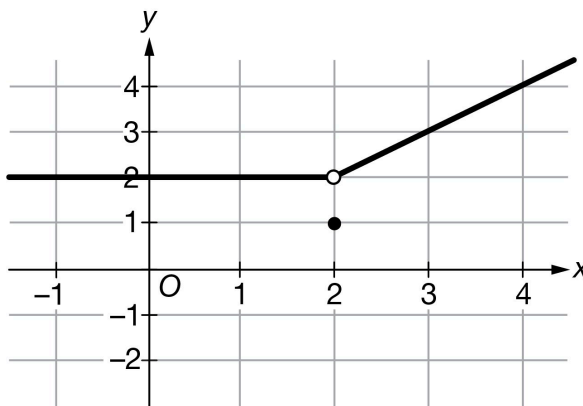
(D) $\lim_{x \rightarrow 6} f(x)$

**Answer B**

Correct. The value of $\lim_{x \rightarrow 6^-} f(x)$, the one-sided limit of f as x approaches 6 from the left, is determined by the part of the graph of f for $x < 6$. As x approaches 6 from the left, the values of $f(x)$ approach 2 along this part of the graph.

Unit 1 AP Topics 1-4

47.

Graph of f

The graph of the function f is shown above. What is $\lim_{x \rightarrow 2} f(x)$?

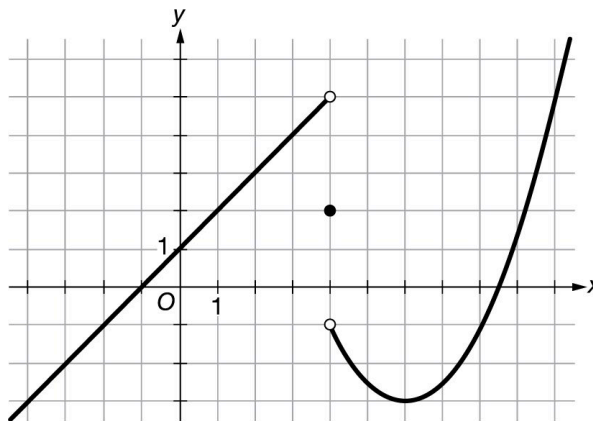
- (A) 0
(B) 1
(C) 2
(D) The limit does not exist.

**Answer C**

Correct. As x approaches 2, the function values approach 2. The fact that $f(2) = 1$ has no bearing on the value of the limit.

Unit 1 AP Topics 1-4

48.

Graph of h

The graph of the function h is shown above. What is $\lim_{x \rightarrow 4} h(x)$?

- (A) -1
(B) 2
(C) 5
(D) nonexistent

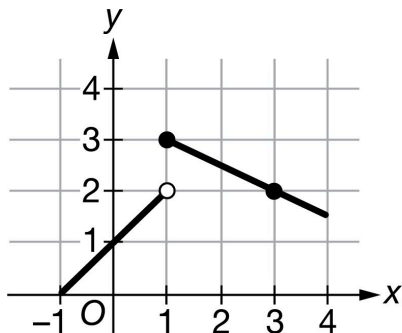
**Answer D**

Correct. Because $\lim_{x \rightarrow 4^-} h(x) \neq \lim_{x \rightarrow 4^+} h(x)$, the limit of $h(x)$ as x approaches 4 does not exist.

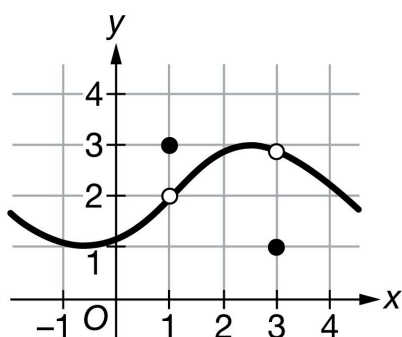
Unit 1 AP Topics 1-4

49. A function f satisfies $\lim_{x \rightarrow 1} f(x) = 3$. Which of the following could be the graph of f ?

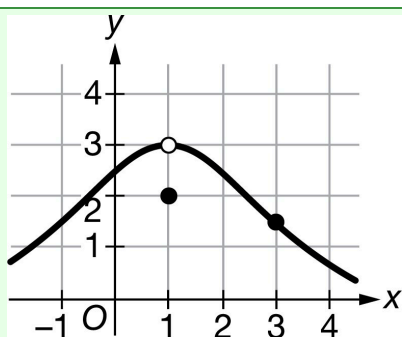
(A)



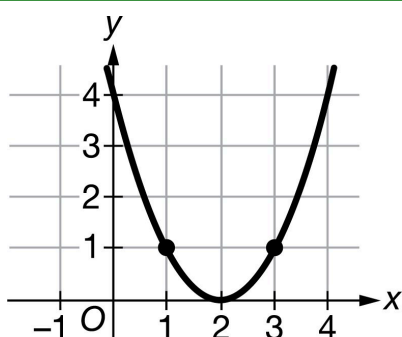
(B)



(C)



(D)

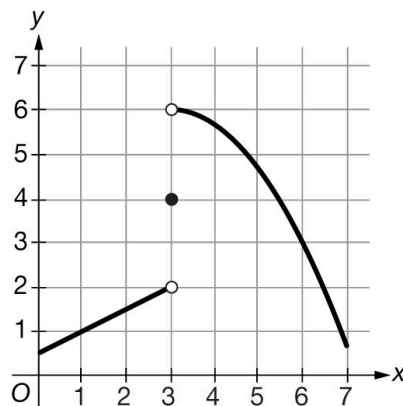


Answer C

Correct. In this graph, the values of $f(x)$ approach 3 as x approaches 1 from the left and as x approaches 1 from the right, that is, $\lim_{x \rightarrow 1^-} f(x) = 3 = \lim_{x \rightarrow 1^+} f(x)$. Since the one-sided limits have the common value 3, $\lim_{x \rightarrow 1} f(x)$ exists and is equal to 3.

Unit 1 AP Topics 1-4

50.

Graph of f

The graph of the function f is shown above. Which of the following expressions equals 2 ?

(A) $f(3)$

(B) $\lim_{x \rightarrow 3^-} f(x)$

(C) $\lim_{x \rightarrow 3^+} f(x)$

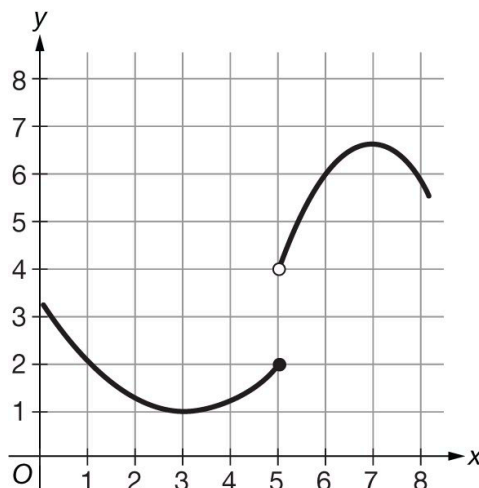
(D) $\lim_{x \rightarrow 3} f(x)$

**Answer B**

Correct. The value of $\lim_{x \rightarrow 3^-} f(x)$, the one-sided limit of f as x approaches 3 from the left, is determined by the part of the graph of f for $x < 3$. As x approaches 3 from the left, the values of $f(x)$ approach 2 along this part of the graph.

Unit 1 AP Topics 1-4

51.

Graph of f

The graph of the function f is shown above. The value of $\lim_{x \rightarrow 5} f(x)$ is

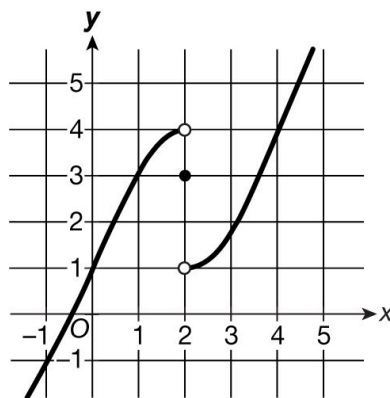
- (A) 2
(B) 3
(C) 4
(D) nonexistent

**Answer D**

Correct. The limit exists if $\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^+} f(x)$. But here $\lim_{x \rightarrow 5^-} f(x) = 2$ and $\lim_{x \rightarrow 5^+} f(x) = 4$, so the limit does not exist.

Unit 1 AP Topics 1-4

52.

Graph of f

The graph of the function f is shown above. What is $\lim_{x \rightarrow 2^+} f(x)$?

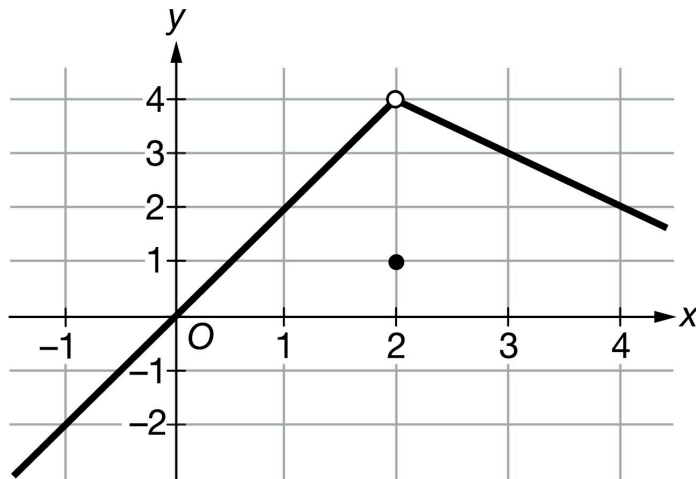
- (A) 1
(B) 3
(C) 4
(D) nonexistent

**Answer A**

Correct. $\lim_{x \rightarrow 2^+} f(x)$ is the limit of $f(x)$ as x approaches 2 from the right. The portion of the graph to the right of $x = 2$ has y -values approaching 1 as the x -values approach 2.

Unit 1 AP Topics 1-4

53.

Graph of f

The graph of the function f is shown above. What is $\lim_{x \rightarrow 2} f(x)$?

(A) $\frac{1}{2}$

(B) 1

(C) 4

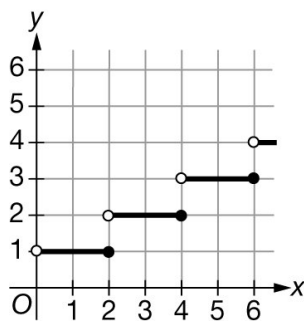
(D) The limit does not exist.

**Answer C**

Correct. As x approaches 2, the function values approach 4. The fact that $f(2) = 1$ has no bearing on the value of the limit.

Unit 1 AP Topics 1-4

54.

Graph of f

The graph of the function f is shown above. The value of $\lim_{x \rightarrow 4} f(x)$ is

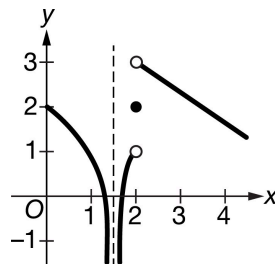
- (A) 2
(B) $\frac{5}{2}$
(C) 3
(D) nonexistent

**Answer D**

Correct. The limit exists if $\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^+} f(x)$. But here $\lim_{x \rightarrow 4^-} f(x) = 2$ and $\lim_{x \rightarrow 4^+} f(x) = 3$, so the limit does not exist.

Unit 1 AP Topics 1-4

55.

Graph of f

The graph of the function f is shown above. What is $\lim_{x \rightarrow 2^+} f(x)$?

(A) 3

(B) 2

(C) 1

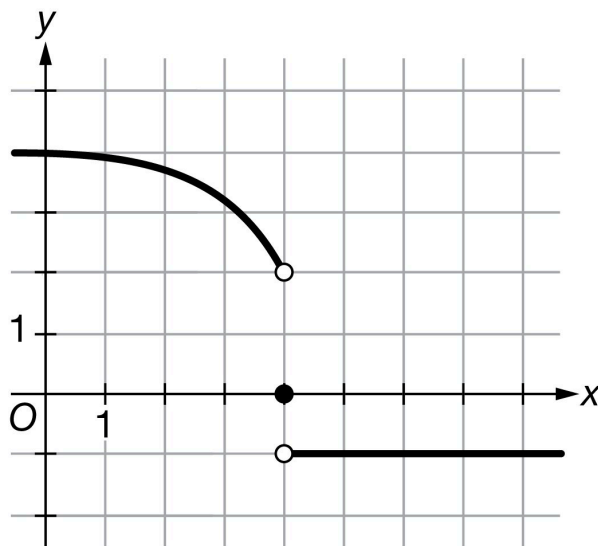
(D) nonexistent

**Answer A**

Correct. $\lim_{x \rightarrow 2^+} f(x)$ is the limit of $f(x)$ as x approaches 2 from the right. The portion of the graph to the right of $x = 2$ has y -values approaching 3 as the x -values approach 2.

Unit 1 AP Topics 1-4

56.

Graph of h

The graph of the function h is shown above. What is $\lim_{x \rightarrow 4} h(x)$?

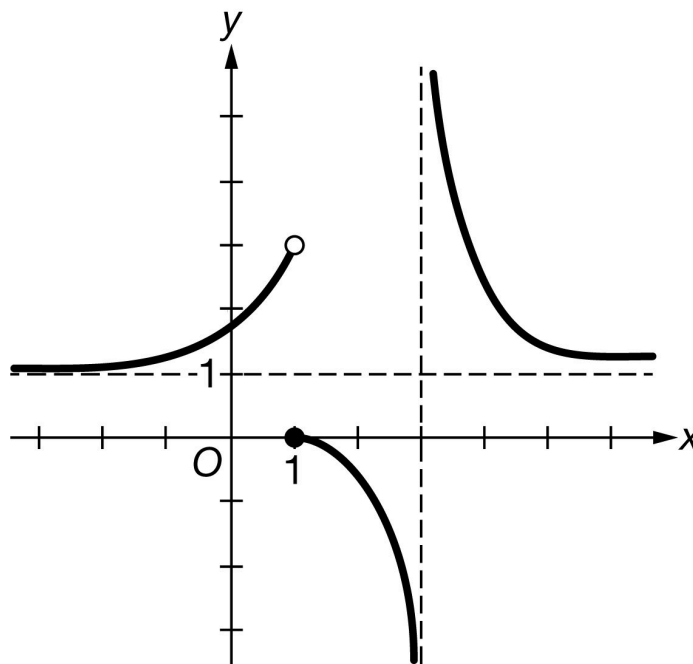
- (A) -1
(B) 0
(C) 2
(D) nonexistent

**Answer D**

Correct. Because $\lim_{x \rightarrow 4^-} h(x) \neq \lim_{x \rightarrow 4^+} h(x)$, the limit of $h(x)$ as x approaches 4 does not exist.

Unit 1 AP Topics 1-4

57.

Graph of f

The graph of a function f is shown above. Which of the following statements is true?

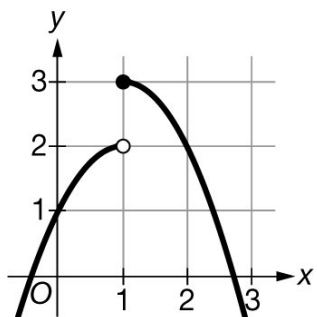
- (A) $\lim_{x \rightarrow 1} f(x) = 1.5$
- (B) $\lim_{x \rightarrow 1} f(x) = 0$
- (C) $\lim_{x \rightarrow 1} f(x)$ does not exist because the left-hand and right-hand limits of $f(x)$ as x approaches 1 do not exist.
- (D) $\lim_{x \rightarrow 1} f(x)$ does not exist because while the left-hand and right-hand limits of $f(x)$ as x approaches 1 exist, their values are not equal. ✓

Answer D

Correct. This statement is true. $\lim_{x \rightarrow 1^-} f(x) = 3 \neq 0 = \lim_{x \rightarrow 1^+} f(x)$; therefore, $\lim_{x \rightarrow 1} f(x)$ does not exist.

Unit 1 AP Topics 1-4

58.

Graph of f

The graph of a function f is shown above. Which of the following statements is true?

- (A) $\lim_{x \rightarrow 1} f(x) = 2.5$
- (B) $\lim_{x \rightarrow 1} f(x) = 3$
- (C) $\lim_{x \rightarrow 1} f(x)$ does not exist because the left-hand and right-hand limits of $f(x)$ as x approaches 1 do not exist.
- (D) $\lim_{x \rightarrow 1} f(x)$ does not exist because while the left-hand and right-hand limits of $f(x)$ as x approaches 1 exist, their values are not equal. ✓

Answer D

Correct. This statement is true. $\lim_{x \rightarrow 1^-} f(x) = 2 \neq 3 = \lim_{x \rightarrow 1^+} f(x)$; therefore, $\lim_{x \rightarrow 1} f(x)$ does not exist.

Unit 1 AP Topics 1-4

59.  The function f is given by $f(x) = 0.2x^4 - 1.0x^3 - 6.6x^2 + 15.4x - 1.99$. For how many positive values of b does $\lim_{x \rightarrow b} f(x) = 6$?

- (A) One
(B) Two
(C) Three
(D) Four

**Answer C**

Correct. Since the polynomial f is a continuous function, $\lim_{x \rightarrow b} f(x)$ is equal to $f(b)$. The number of positive values of b for which the limit of $f(x)$ is equal to 6 can therefore be found by examining the graphs of $y = f(x)$ and $y = 6$ to see how many times they intersect. Care must be taken, however, since graphical representations of functions may miss important function behavior because of issues of scale. For example, a window of $0 \leq x \leq 10$ and $-60 \leq y \leq 250$ indicates that there is a value of b near $x = 8$ where the graph of $y = f(x)$ crosses the line $y = 6$. The scale along the y -axis is too large, however, to discern the behavior of f for values of x near 0. Even using the interval $0 \leq y \leq 8$, the scale may suggest that the graph of $y = f(x)$ is tangent to the line $y = 6$ near $x = 1$. Careful zooming, however, around $x = 1$ and $y = 6$ will show that the graph of $y = f(x)$ actually crosses the line $y = 6$ twice. Therefore, there are three positive values of b for which $f(b) = 6$. The remaining solution to $f(b) = 6$ is for a value of b that is negative.

Unit 1 AP Topics 1-4

60.  The function f is given by $f(x) = 0.1x^4 - 0.5x^3 - 3.3x^2 + 7.7x - 1.99$. For how many positive values of b does $\lim_{x \rightarrow b} f(x) = 2$?

- (A) One
(B) Two
(C) Three
(D) Four

**Answer C**

Correct. Since the polynomial f is a continuous function, $\lim_{x \rightarrow b} f(x)$ will be equal to $f(b)$. The number of positive values of b for which the limit of $f(x)$ is equal to 2 can therefore be found by examining the graphs of $y = f(x)$ and $y = 2$ to see how many times they intersect. Care must be taken, however, since graphical representations of functions may miss important function behavior because of issues of scale. For example, a window of $0 \leq x \leq 10$ and $-60 \leq y \leq 250$ indicates that there is a value of b near $x = 8$ where the graph of $y = f(x)$ crosses the line $y = 2$. The scale along the y -axis is too large, however, to discern the behavior of f for values of x near 0. Even using the interval $0 \leq y \leq 4$, the scale may suggest that the graph of $y = f(x)$ is tangent to the line $y = 2$ near $x = 1$. Careful zooming, however, around $x = 1$ and $y = 2$ will show that the graph of $y = f(x)$ actually crosses the line $y = 2$ twice. Therefore there are three positive values of b for which $f(b) = 2$. The remaining solution to $f(b) = 2$ is for $b < 0$.

Unit 1 AP Topics 1-4

61. Of the following tables, which best reflects the values of a function g for which $\lim_{x \rightarrow 9} g(x) = 5$?

(A)

x	4.85	4.90	4.95	4.99	5.01	5.05	5.10	5.15
$g(x)$	8.700	8.800	8.900	8.980	9.020	9.100	9.200	9.300

(B)

x	8.85	8.90	8.95	8.99	9.01	9.05	9.10	9.15
$g(x)$	4.925	4.950	4.975	4.995	5.005	5.025	5.050	5.075



(C)

x	8.85	8.90	8.95	8.99	9.01	9.05	9.10	9.15
$g(x)$	4.950	4.967	4.983	4.997	10.030	10.070	10.120	10.170

(D)

x	8.85	8.90	8.95	8.99	9.01	9.05	9.10	9.15
$g(x)$	6.667	10.000	20.000	100.000	4.570	4.590	4.615	4.640

Answer B

Correct. The values of $g(x)$ in this table are increasing toward 5 as x approaches 9 from the left, and they are also increasing toward 5 as x approaches 9 from the right. The table therefore suggests that $\lim_{x \rightarrow 9^-} g(x) = 5 = \lim_{x \rightarrow 9^+} g(x)$ and hence that $\lim_{x \rightarrow 9} g(x) = 5$.

Unit 1 AP Topics 1-4

62.

x	1.9	1.99	1.999	1.9999	2.0001	2.001	2.01	2.1
$f(x)$	9.80	9.85	9.90	9.95	9.95	9.90	9.85	9.80

The table above gives values of the function f at selected values of x . Which of the following conclusions is supported by the data in the table?

(A) $\lim_{x \rightarrow 2} f(x) = 10$



(B) $\lim_{x \rightarrow 2} f(x) = \infty$

(C) $\lim_{x \rightarrow 10} f(x) = 2$

(D) $\lim_{x \rightarrow 10} f(x) = \infty$

Answer A

Correct. The data in the table suggest that as x approaches 2, the values of $f(x)$ approach 10.

63.

x	3.9	3.99	3.999	3.9999	4.0001	4.001	4.01	4.1
$f(x)$	5.8	5.85	5.9	5.95	6.999	6.99	6.9	6

The table above gives values of the function f at selected values of x . Which of the following conclusions is supported by the data in the table?

(A) $\lim_{x \rightarrow 4} f(x) = 6$

(B) $\lim_{x \rightarrow 4} f(x) = 7$

(C) $\lim_{x \rightarrow 4^-} f(x) = 6$ and $\lim_{x \rightarrow 4^+} f(x) = 7$



(D) $\lim_{x \rightarrow 4^-} f(x) = 7$ and $\lim_{x \rightarrow 4^+} f(x) = 6$

Answer C

Correct. The data in the table suggest that as x approaches 4 from the left, the values of $f(x)$ approach 6. The data in the table also suggest that as x approaches 4 from the right, the values of $f(x)$ approach 7.

Unit 1 AP Topics 1-4

64.

x	10	10.9	10.99	10.999	11.001	11.01	11.1	12
$f(x)$	29	31.7	31.97	31.997	32.003	32.03	32.3	35

The table above gives values of the function f at selected values of x . Which of the following conclusions is supported by the data in the table?

(A) $\lim_{x \rightarrow 11} f(x) = 32$ ✓

(B) $\lim_{x \rightarrow 11} f(x) = \infty$

(C) $\lim_{x \rightarrow 32} f(x) = 11$

(D) $\lim_{x \rightarrow 32} f(x) = \infty$

Answer A

Correct. The data in the table suggest that as x approaches 11, the values of $f(x)$ approach 32.

65.

x	2	2.9	2.99	2.999	3.001	3.01	3.1	4
$f(x)$	-8	-80	-800	-8000	8000	800	80	8

The table above gives values of a function f at selected values of x . Which of the following conclusions is supported by the data in the table?

(A) $\lim_{x \rightarrow 3} f(x) = 0$

(B) $\lim_{x \rightarrow 3} f(x) = 3$

(C) $\lim_{x \rightarrow 3} f(x) = 10$

(D) $\lim_{x \rightarrow 3} f(x)$ does not exist. ✓

Answer D

Correct. The data in the table suggest that as x approaches 3 from the left, the values of $f(x)$ increase. The data in the table also suggest that as x approaches 3 from the right, the values of $f(x)$ decrease. Because there is no evidence to suggest that $f(x)$ approaches a fixed value as x approaches 3, a reasonable conclusion is that the limit does not exist.

Unit 1 AP Topics 1-4

66.

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$	7.018	7.007	7.002	6.998	5.982	5.887

The table above gives selected values for a continuous function f . Based on the data in the table, which of the following is the best approximation for $\lim_{x \rightarrow 4} f(x)$?

(A) 0

(B) 4

(C) 7



(D) There is no best approximation, because the limit does not exist.

Answer C

Correct. As x approaches 4 from the left and from the right, it appears that the values of $f(x)$ are getting closer to 7 based on the information in the table.

67.

x	4.9	4.99	4.999	4.9999	5.0001	5.001	5.01	5.1
$f(x)$	4	-16	-256	-726	7.9999	7.999	7.99	7.9

The table above gives values of a function f at selected values of x . Which of the following conclusions is supported by the data in the table?

(A) $\lim_{x \rightarrow 5} f(x) = 8$ (B) $\lim_{x \rightarrow 5^-} f(x) = 8$ (C) $\lim_{x \rightarrow 5^+} f(x) = 8$ (D) $\lim_{x \rightarrow 8^+} f(x) = 5$

Answer C

Correct. The data in the table suggest that as x approaches 5 from the right, the values of $f(x)$ approach 8.

Unit 1 AP Topics 1-4

68.

x	1.9	1.99	1.999	1.9999	2	2.0001	2.001	2.01	2.1
$f(x)$	7.80	7.86	7.90	7.95	3	8.05	8.10	8.14	8.20

The table above gives values of the function f at selected values of x . Which of the following statements must be true?

- (A) $\lim_{x \rightarrow 2} f(x) = 3$
- (B) $\lim_{x \rightarrow 2} f(x) = 8$
- (C) $\lim_{x \rightarrow 2} f(x)$ does not exist.
- (D) $\lim_{x \rightarrow 2} f(x)$ cannot be definitively determined from the data in the table. ✓

Answer D

Correct. The values of a function at a finite set of points may approximate but cannot definitely determine the value of the limit of the function at a particular point or whether the limit exists at a particular point.

69.

x	0.9	0.99	0.999	0.9999	1	1.0001	1.001	1.01	1.1
$f(x)$	6.80	6.86	6.90	6.95	2	7.05	7.10	7.14	7.20

The table above gives values of the function f at selected values of x . Which of the following statements must be true?

- (A) $\lim_{x \rightarrow 1} f(x) = 2$
- (B) $\lim_{x \rightarrow 1} f(x) = 7$
- (C) $\lim_{x \rightarrow 1} f(x)$ does not exist.
- (D) $\lim_{x \rightarrow 1} f(x)$ cannot be definitively determined from the data in the table. ✓

Answer D

Correct. The values of a function at a finite set of points cannot determine the value of the limit of the function at a particular point or even determine whether the limit exists at a particular point.

Unit 1 AP Topics 1-4

70. Of the following tables, which best reflects the values of a function g for which $\lim_{x \rightarrow 7} g(x) = 6$?

(A)

x	5.85	5.90	5.95	5.99	6.01	6.05	6.10	6.15
$g(x)$	7.126	7.075	7.033	7.006	6.995	6.977	6.964	6.960

(B)

x	6.85	6.90	6.95	6.99	7.01	7.05	7.10	7.15
$g(x)$	5.620	5.837	5.961	5.998	5.999	5.964	5.863	5.709



(C)

x	6.85	6.90	6.95	6.99	7.01	7.05	7.10	7.15
$g(x)$	5.919	5.942	5.969	5.993	7.017	7.087	7.177	7.269

(D)

x	6.85	6.90	6.95	6.99	7.01	7.05	7.10	7.15
$g(x)$	1.362	5.954	10.691	14.690	6.010	6.049	6.095	6.140

Answer B

Correct. The values of $g(x)$ in this table are increasing toward 6 as x approaches 7 from the left, and they are also increasing toward 6 as x approaches 7 from the right. The table therefore suggests that $\lim_{x \rightarrow 7^-} g(x) = 6 = \lim_{x \rightarrow 7^+} g(x)$ and hence that $\lim_{x \rightarrow 7} g(x) = 6$.

Unit 1 AP Topics 1-4

71.

x	3.9	3.99	3.999	3.9999	4.0001	4.001	4.01	4.1
$f(x)$	5	-25	125	-625	5.9999	5.999	5.99	5.9

The table above gives values of a function f at selected values of x . Which of the following conclusions is supported by the data in the table?

(A) $\lim_{x \rightarrow 4} f(x) = 6$

(B) $\lim_{x \rightarrow 4^-} f(x) = 6$

(C) $\lim_{x \rightarrow 4^+} f(x) = 6$

(D) $\lim_{x \rightarrow 6^+} f(x) = 4$

**Answer C**

Correct. The data in the table suggest that as x approaches 4 from the right, the values of $f(x)$ approach 6.

72.

x	2.9	2.99	2.999	3.001	3.01	3.1
$f(x)$	5.018	5.007	5.002	4.998	4.982	4.887

The table above gives selected values for a continuous function f . Based on the data in the table, which of the following is the best approximation for $\lim_{x \rightarrow 3} f(x)$?

(A) 0

(B) 3

(C) 5

(D) There is no best approximation, because the limit does not exist.

**Answer C**

Correct. As x approaches 3 from the left and from the right, it appears that the values of $f(x)$ are getting closer to 5 based on the information in the table.

Unit 1 AP Topics 1-4

73.

x	1	1.9	1.99	1.999	1.9999	2.0001	2.001	2.01	2.1	3
$f(x)$	-4	-1.399	-1.040	-1.004	-1.000	6.001	6.012	6.121	7.261	25

The table above gives values of the function f at selected values of x . Which of the following conclusions is supported by the data in the table?

(A) $\lim_{x \rightarrow 2} f(x) = -1$

(B) $\lim_{x \rightarrow 2} f(x) = 6$

(C) $\lim_{x \rightarrow 2^-} f(x) = -1$ and $\lim_{x \rightarrow 2^+} f(x) = 6$ ✓

(D) $\lim_{x \rightarrow 2^-} f(x) = 6$ and $\lim_{x \rightarrow 2^+} f(x) = -1$

Answer C

Correct. The data in the table suggest that as x approaches 2 from the left, the values of $f(x)$ approach -1 . The data in the table also suggest that as x approaches 2 from the right, the values of $f(x)$ approach 6.

Unit 1 AP Topics 1-4

74.

x	5	5.9	5.99	5.999	6.001	6.01	6.1	7
$f(x)$	2	20	200	2000	-2000	-200	-20	-2

The table above gives values of a function f at selected values of x . Which of the following conclusions is supported by the data in the table?

(A) $\lim_{x \rightarrow 6} f(x) = 0$

(B) $\lim_{x \rightarrow 6} f(x) = 6$

(C) $\lim_{x \rightarrow 6} f(x) = 10$

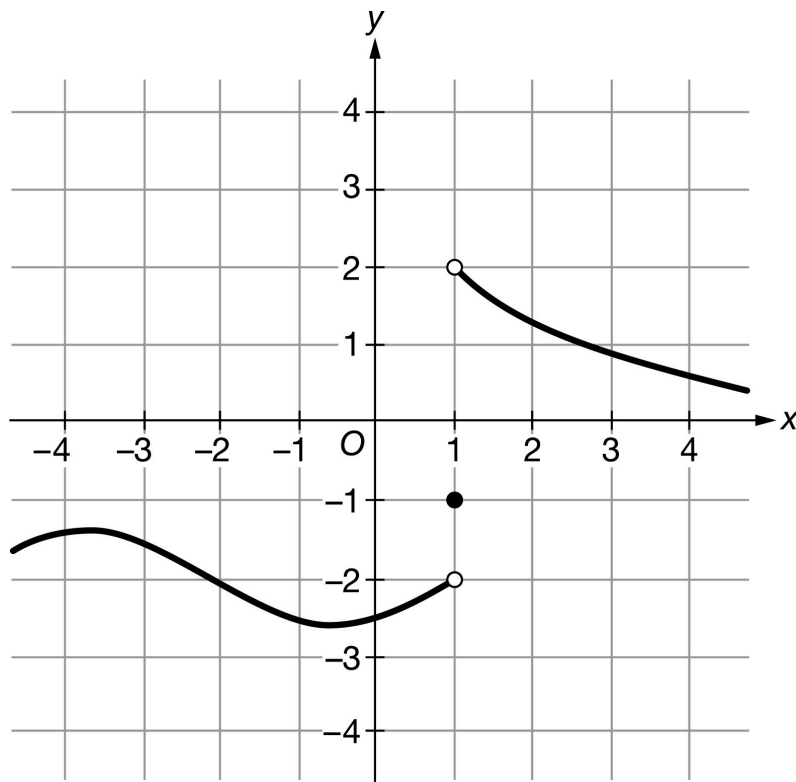
(D) $\lim_{x \rightarrow 6} f(x)$ does not exist.

**Answer D**

Correct. The data in the table suggest that as x approaches 6 from the left, the values of $f(x)$ increase. The data in the table also suggest that as x approaches 6 from the right, the values of $f(x)$ decrease. Because there is no evidence to suggest that $f(x)$ approaches a fixed value as x approaches 6, a reasonable conclusion is that the limit does not exist.

Unit 1 AP Topics 1-4

75.

Graph of f

The graph of the function f is shown in the figure above. The value of $\lim_{x \rightarrow 1^+} f(x)$ is

- (A) -2
- (B) -1
- (C) 2
- (D) nonexistent

**Answer C**

Unit 1 AP Topics 1-4

76. Let f be a function that is defined for all real numbers x . Of the following, which is the best interpretation of the statement $\lim_{x \rightarrow 3} f(x) = 5$?
- (A) The value of the function f at $x = 3$ is 5.
 - (B) The value of the function f at $x = 5$ is 3.
 - (C) As x approaches 3, the values of $f(x)$ approach 5. ✓
 - (D) As x approaches 5, the values of $f(x)$ approach 3.

Answer C

Correct. This is a good, informal description of the meaning of the mathematical statement.

77. The function f has the property that as x gets closer and closer to 4, the values of $f(x)$ get closer and closer to 7. Which of the following statements must be true?
- (A) $f(4) = 7$
 - (B) $f(7) = 4$
 - (C) $\lim_{x \rightarrow 4} f(x) = 7$ ✓
 - (D) $\lim_{x \rightarrow 7} f(x) = 4$

Answer C

Correct. The given information about the function f is a good, informal description of the mathematical statement $\lim_{x \rightarrow 4} f(x) = 7$.

Unit 1 AP Topics 1-4

78. Let f be the function given by $f(x) = \frac{e^{2x}-1}{x}$. Which of the following equations expresses the property that $f(x)$ can be made arbitrarily close to 2 by taking x sufficiently close to 0, but not equal to 0?
- (A) $f(0) = 2$
- (B) $f\left(\lim_{x \rightarrow 0} x\right) = 2$
- (C) $\lim_{x \rightarrow 0} f(x) = 2$ ✓
- (D) $\lim_{x \rightarrow 2} f(x) = 0$

Answer C

Correct. This is the correct notation that the limit of $f(x)$ is 2 as x approaches 0 (but is not equal to 0 where f is undefined).

79. The function f has the property that as x gets closer and closer to 3, the values of $f(x)$ get closer and closer to 5. Which of the following statements must be true?
- (A) $f(3) = 5$
- (B) $f(5) = 3$
- (C) $\lim_{x \rightarrow 3} f(x) = 5$ ✓
- (D) $\lim_{x \rightarrow 5} f(x) = 3$

Answer C

Correct. The given information about the function f is a good, informal description of the mathematical statement $\lim_{x \rightarrow 3} f(x) = 5$.

Unit 1 AP Topics 1-4

80. Let f be a function that is defined for all real numbers x . Of the following, which is the best interpretation of the statement $\lim_{x \rightarrow 2} f(x) = 7$?
- (A) The value of the function f at $x = 2$ is 7.
- (B) The value of the function f at $x = 7$ is 2.
- (C) As x approaches 2, the values of $f(x)$ approach 7. ✓
- (D) As x approaches 7, the values of $f(x)$ approach 2.

Answer C

Correct. This is a good, informal description of the meaning of the mathematical statement.

81. Let f be the function given by $f(x) = \frac{e^{3x}-1}{x}$ for $x \neq 0$. Which of the following equations expresses the property that $f(x)$ can be made arbitrarily close to 3 by taking x sufficiently close to 0, but not equal to 0?
- (A) $f(0) = 3$
- (B) $f\left(\lim_{x \rightarrow 0} x\right) = 3$
- (C) $\lim_{x \rightarrow 0} f(x) = 3$ ✓
- (D) $\lim_{x \rightarrow 3} f(x) = 0$

Answer C

Correct. This is the correct notation that the limit of $f(x)$ is 3 as x approaches 0 (but is not equal to 0 where f is undefined).

Unit 1 AP Topics 1-4

82. Let $v(t)$ be the velocity, in feet per second, of a skydiver at time t seconds, $t \geq 0$. After her parachute opens, her velocity satisfies the differential equation $\frac{dv}{dt} = -2v - 32$, with initial condition $v(0) = -50$.
- (a) Use separation of variables to find an expression for v in terms of t , where t is measured in seconds.
- (b) Terminal velocity is defined as $\lim_{t \rightarrow \infty} v(t)$. Find the terminal velocity of the skydiver to the nearest foot per second.
- (c) It is safe to land when her speed is 20 feet per second. At what time t does she reach this speed?

Part A

Solution: Max 6/6

1 point is earned for correctly separating variables

$$\frac{dv}{dt} = -2v - 32 = -2(v + 16)$$

1 point is earned for the correct antiderivative of dv side

$$0/1 \text{ if not } \int \frac{dv}{av + b}, a \neq 0$$

$$\frac{dv}{v + 16} = -2dt$$

1 point is earned for the correct antiderivative of dt side

$$\ln |v + 16| = -2t + A$$

1 point is earned for the correct constant of integration

$$|v + 16| = e^{-2t+A} = e^A e^{-2t}$$

1 point is earned for correctly using initial condition $v(0) = -50$

$$v + 16 = Ce^{-2t}$$

$$-50 + 16 = Ce^0; C = -34$$

1 point is earned for correctly solving for $v(t)$

Unit 1 AP Topics 1-4

0/1 if not solving $\frac{dv}{av+b} = k dt$ where a, b, k nonzero

0/1 if no constant of integration

0/6 if variables not separated

$$v = -34e^{-2t} - 16$$

OR

Integrating Factor Solution: Max 5/6

1 point is earned for correct use of integrating factor

$$v'(t) = -2v(t) - 32$$

$$e^{2t}(v'(t) + 2v(t)) = -32e^{2t}$$

$$\frac{d}{dt}(v(t)e^{2t}) = -32e^{2t}$$

$$v(t)e^{2t} = \int -32e^{2t} dt = -16e^{2t} + C$$

$$v(t) = -16 + Ce^{-2t}$$

$$-50 = -16 + C; \quad C = -34$$

4 points are earned for the correct solution

$$v(t) = -16 - 34e^{-2t}$$



0	1	2	3	4	5	6
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The student response earns all of the following points:

Solution: Max 6/6

1 point is earned for correctly separating variables

$$\frac{dv}{dt} = -2v - 32 = -2(v + 16)$$

1 point is earned for the correct antiderivative of dv side

Unit 1 AP Topics 1-4

$$0/1 \text{ if not } \int \frac{dv}{av + b}, a \neq 0$$

$$\frac{dv}{v + 16} = -2dt$$

1 point is earned for the correct antiderivative of dt side

$$\ln |v + 16| = -2t + A$$

1 point is earned for the correct constant of integration

$$|v + 16| = e^{-2t+A} = e^A e^{-2t}$$

1 point is earned for correctly using initial condition $v(0) = -50$

$$v + 16 = Ce^{-2t}$$

$$-50 + 16 = Ce^0; C = -34$$

1 point is earned for correctly solving for $v(t)$

$$0/1 \text{ if not solving } \frac{dv}{av + b} = k dt \text{ where } a, b, k \text{ nonzero}$$

0/1 if no constant of integration

0/6 if variables not separated

$$v = -34e^{-2t} - 16$$

OR

Integrating Factor Solution: Max 5/6

1 point is earned for correct use of integrating factor

Unit 1 AP Topics 1-4

$$v'(t) = -2v(t) - 32$$

$$e^{2t}(v'(t) + 2v(t)) = -32e^{2t}$$

$$\frac{d}{dt}(v(t)e^{2t}) = -32e^{2t}$$

$$v(t)e^{2t} = \int -32e^{2t} dt = -16e^{2t} + C$$

$$v(t) = -16 + Ce^{-2t}$$

$$-50 = -16 + C; \quad C = -34$$

4 points are earned for the correct solution

$$v(t) = -16 - 34e^{-2t}$$

Part B

1 point is earned for the correct limit value must be exponential $v(t)$ with finite limit

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} (-34e^{-2t} - 16) = -16$$



0

1

The student response earns all of the following points:

1 point is earned for the correct limit value must be exponential $v(t)$ with finite limit

$$\lim_{t \rightarrow \infty} v(t) = \lim_{t \rightarrow \infty} (-34e^{-2t} - 16) = -16$$

Part C

1 point is earned for correctly setting $v(t) = -20$

$$v(t) = -34e^{-2t} - 16 = -20$$

1 point is earned for the correct solution must be exponential $v(t)$

Unit 1 AP Topics 1-4

$$e^{-2t} = \frac{2}{17}; \quad t = -\frac{1}{2} \ln \left(\frac{2}{17} \right) = 1.070$$



0	1	2
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The student response earns all of the following points:

1 point is earned for correctly setting $v(t) = -20$

$$v(t) = -34e^{-2t} - 16 = -20$$

1 point is earned for the correct solution must be exponential $v(t)$

$$e^{-2t} = \frac{2}{17}; \quad t = -\frac{1}{2} \ln \left(\frac{2}{17} \right) = 1.070$$